ACE-Step XL 1.5 Premium v5.2 Written Tutorial

This README is the full Markdown tutorial generated from the public YouTube tutorial, local file `ACESTEP_XL_15_Full_Tutorial.mp4`, `captions (2).srt`, live local ACE-Step app screenshots, one newly generated demo song, one Audio Processing run made from that song, and selected 4K frames extracted from the source video.

Generated demo song: `G:\ACE_Step_v1\ACE-Step_Premium\outputs\0021\8642a822-c4d0-ff4a-531f-6d5cfb334381.mp3`

Processed demo output: `G:\ACE_Step_v1\ACE-Step_Premium\outputs\audio_processing_0004\8642a822-c4d0-ff4a-531f-6d5cfb334381_processed.wav`

Source Video, Thumbnail, Links, And Chapters

The public video description presents this as a full ACE-Step XL 1.5 Premium guide for local AI music generation, remix, repaint, stem extraction, audio processing, SAM Audio segmentation, Windows installation, RunPod, Massed Compute, SimplePod, and Linux/cloud workflows.

Tutorial video thumbnail

- **Tutorial video:** Forget Suno: Run the Ultimate AI Music Studio LOCALLY (100% Free)
- **Channel:** SECourses
- **App/latest zip and Patreon post:** Patreon ACESTEP XL Premium SAM Audio post
- **Windows requirements guide:** Windows requirements guide
- **Discord/help/community:** SECourses Discord server
- **Duration:** 1:15:09
- **Published:** June 17, 2026

Video Chapters

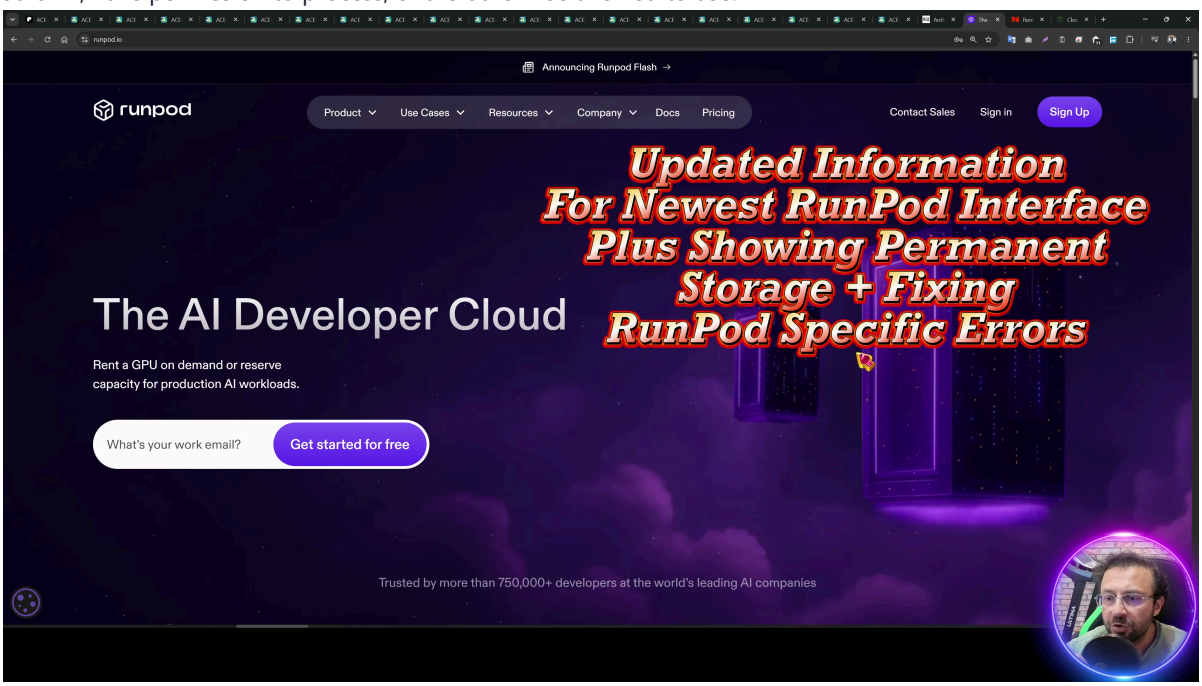
- 0:00 - Intro: ACESTEP XL 1.5 Premium local music, segmentation and processing tutorial
- 0:52 - Fast song generation examples across styles in under one minute
- 1:55 - Output manifest proof, 40-second generation time and supported models
- 2:29 - Turbo/SFT/Base models, LoRA support, GPU presets and Torch Compile boost
- 3:10 - Remix feature preview, same-lyrics requirement and responsible usage note
- 4:16 - Repaint mode: regenerate and merge only a selected song section
- 5:38 - Extract mode: stems, silence trimming, all-stems and batch folders
- 6:30 - Lego mode: add an instrument stem such as guitar into existing audio
- 7:25 - Audio Processing presets and manual enhancement controls for AI songs
- 8:35 - Auto-Editor silent trim for tutorials, videos, audio and workflow export
- 9:48 - DaVinci/Premiere/Final Cut/ShotCut/Kdenlive timeline export demo
- 11:01 - SAM Audio Segment: BF16 models, VRAM presets and advanced segmentation
- 11:47 - SAM outputs demo: vocals, drums, bass, remaining audio and saved files
- 12:47 - Custom SAM prompts, semicolon batch segmenting and speech cleanup example
- 14:19 - Batch processing, load metadata, manifests, saved settings and presets
- 15:09 - Why local open-source models matter and where to run ACESTEP
- 15:55 - Windows install begins: Patreon zip, changelog, attachments and download
- 16:53 - Windows requirements tutorial before Python/CUDA/C++/FFmpeg setup
- 17:29 - Extract zip safely, avoid bad paths and run Windows_Install_or_Update.bat
- 18:24 - Automatic VENV, FFmpeg, UV install, model downloads and hash verification
- 19:24 - Turbo default vs all-model download for SFT/Base and BF16 safetensors
- 20:32 - First Windows launch, default Generate Song test and CMD progress
- 21:44 - Model recommendations, VRAM tiers, languages, vocals and MP4 image output

- 23:29 - Torch Compile setup for faster repeated generations
- 24:05 - Outputs folder, model switching and full remix setup workflow
- 25:24 - Practical remix loop: adapted lyrics, strength, reference audio and seed lock
- 28:03 - Repaint workflow with source range preview, generated result and comparison
- 29:13 - Recap: extraction, Lego, audio processing and SAM text-prompt usage
- 30:20 - Windows wrap-up, LoRA training teaser and move to cloud installs
- 31:16 - RunPod setup: credits, template, CUDA filters, GPU choice and storage
- 34:53 - Upload zip in Jupyter Lab, extract, run instructions and handle installs
- 35:43 - RunPod errors, resume behavior, model downloads and hash verification
- 38:04 - Start ACESTEP on RunPod with Gradio Live, proxy ports and persistence
- 40:18 - Add 7860/7861 ports, verify storage reuse and rerun installer after resume
- 42:10 - RunPod connection troubleshooting and Gradio Live recommendation
- 44:12 - Fix corrupted VENV/stale handle errors, reinstall safely and retest
- 47:24 - Successful RunPod relaunch, default generation, nvitop and loading tips
- 49:26 - RunPod first load vs fast inference, 15-second second generation example
- 51:02 - Download outputs and delete RunPod pods/storage to stop spending
- 53:30 - Massed Compute setup: coupon, Creator image, GPU prices and ThinLinc
- 57:13 - Massed install from extracted folder, Linux notes and ultra-fast downloads
- 59:18 - Start app on Massed Compute via localhost or Gradio Live
- 1:00:23 - Default Massed generation, nvitop, faster loading and speed test
- 1:02:03 - Sync/download outputs and delete Massed Compute instance safely
- 1:03:25 - SimplePod setup: template, persistent volume, pricing and GPU choice
- 1:06:39 - Jupyter upload, direct file browser, install command and model downloads
- 1:08:21 - Start SimplePod, Gradio Live, default generation and one-time load errors
- 1:09:31 - nvitop monitoring, newer driver/CUDA details and generation completion
- 1:10:42 - Direct output/model downloads through SimplePod file browser
- 1:11:42 - Delete instance, keep storage, relaunch GPU and verify install
- 1:13:15 - Discord, subreddit, changelog, update guidance and support links
- 1:14:30 - Final cleanup: terminate servers, delete storage and LoRA training outro

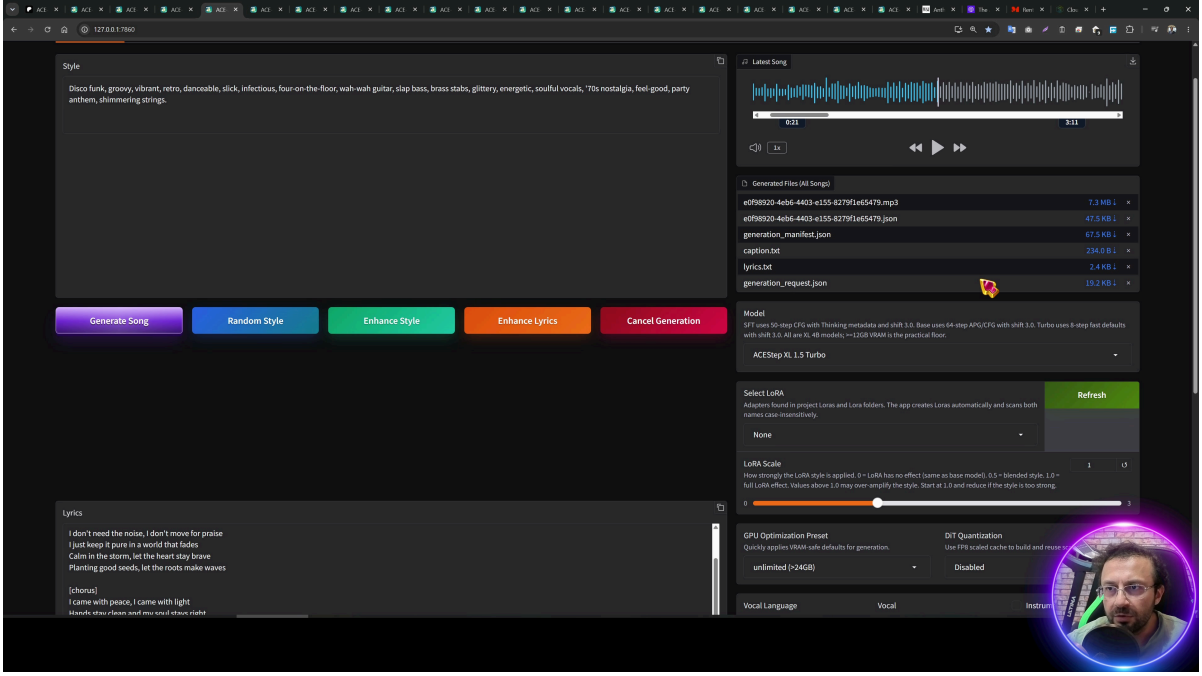
1. What ACE-Step XL 1.5 Premium Is

ACE-Step XL 1.5 Premium is a local-first music generation and audio utility suite. The video presents it as more than a song generator: it also includes advanced generation modes, remix/repaint workflows, stem extraction, LEGO-style stem addition, SAM Audio segmentation, Auto-Editor trimming, mastering-style audio processing, dataset tools, and LoRA/LoKr training pages.

Responsible-use note: the source tutorial says to use the application respectfully and for research/education. For remix, repaint, extraction, and pitch work, use material you own, have permission to process, or are otherwise allowed to use.



Video introduction



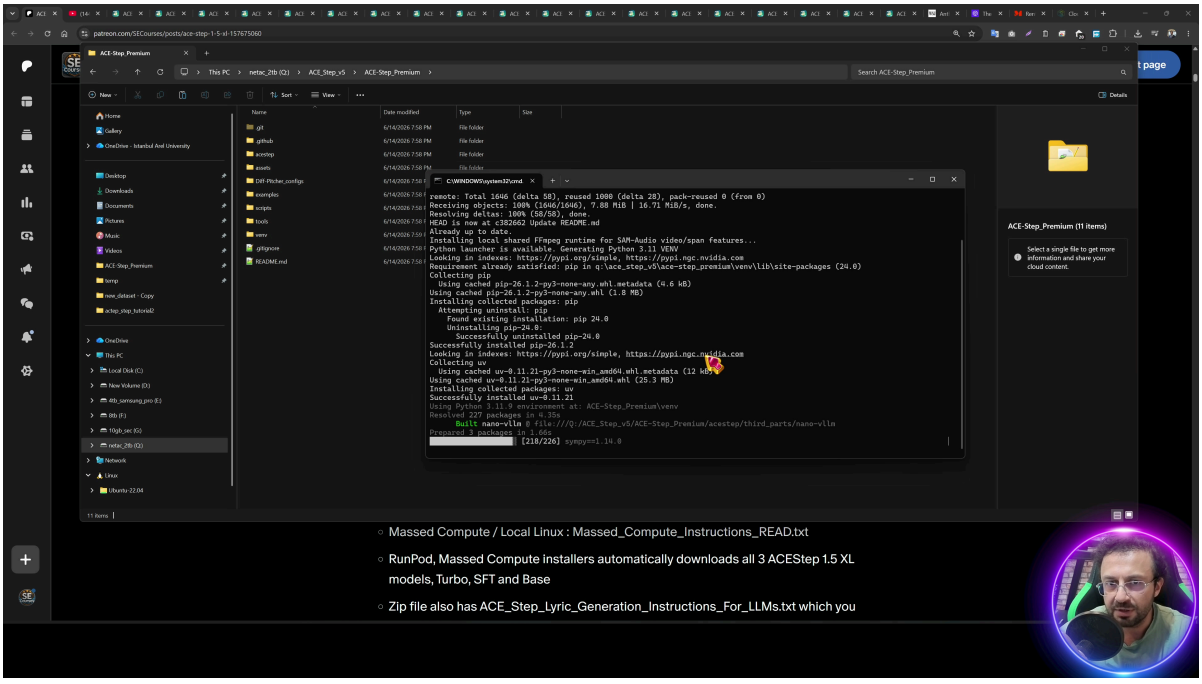
Feature overview

Core jobs covered in the tutorial:

- Generate complete songs from a style prompt, structured lyrics, model choice, duration, language, and seed settings.
- Remix, repaint, extract, LEGO-add, complete, retake, edit, and reuse LM code hints when the selected model supports the workflow.
- Trim silence, export audio/video, enhance or pre-master generated songs, and optionally run pitch correction with DiffPitcher.
- Use SAM Audio with quick prompts, custom prompts, explicit spans, or batch prompt lists to extract target audio and save the remaining audio.
- Use Library, Load Metadata, Results, and the preset system to restore, inspect, score, save, and reuse generation runs.

2. Install And Start On Windows

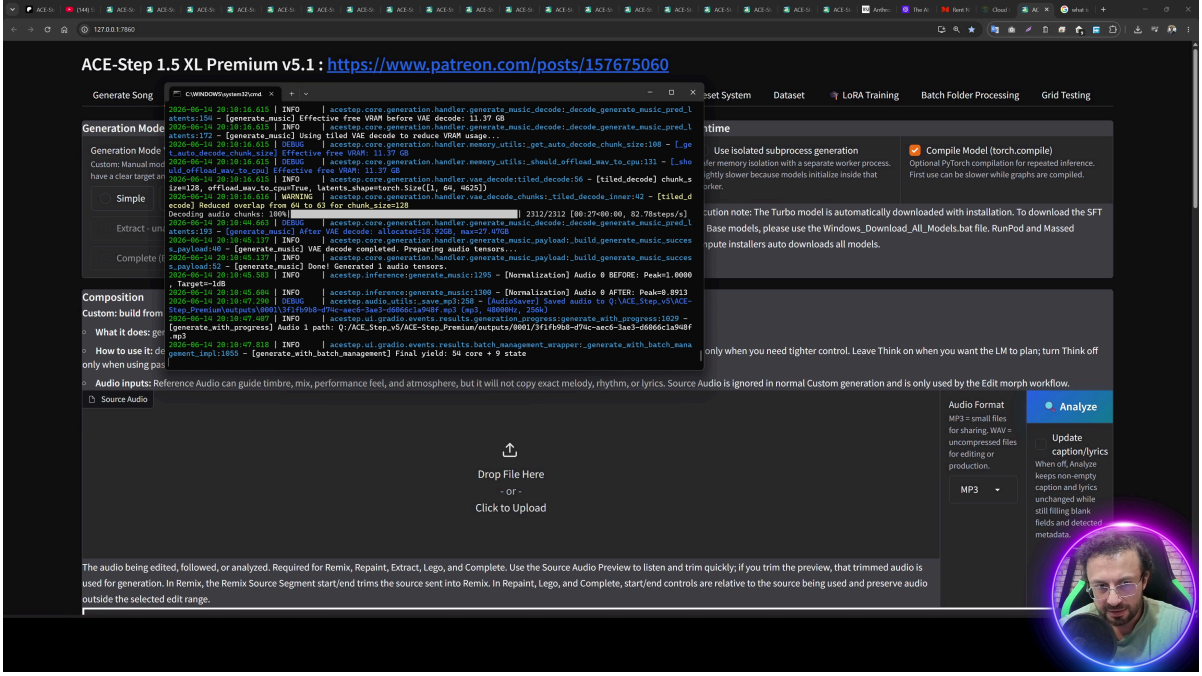
The Windows workflow uses the included batch files. Extract the ZIP, keep the folder structure intact, run the installer/update script, optionally download all models, then start the app with the Windows launcher.



Windows installer

1. Extract the ACE-Step Premium ZIP to a path with enough free disk space for the virtual environment, model files, outputs, and FFmpeg runtime.
2. Run `Windows_Install_or_Update.bat`. The installer creates the Python virtual environment, downloads or uses shared FFmpeg, installs packages with UV, and prepares the app.
3. Run `Windows_Download_All_Models.bat` if you want SFT and Base models in addition to the automatically available Turbo path.
4. Run `Windows_Start_App.bat`. In this workspace the launcher started ACE-Step at `http://127.0.0.1:7862` because other Gradio apps were already using `7860` and `7861`.
5. Watch the command window for model download, model load, generation, and error details. The video recommends trusting the terminal status more than only the browser UI.

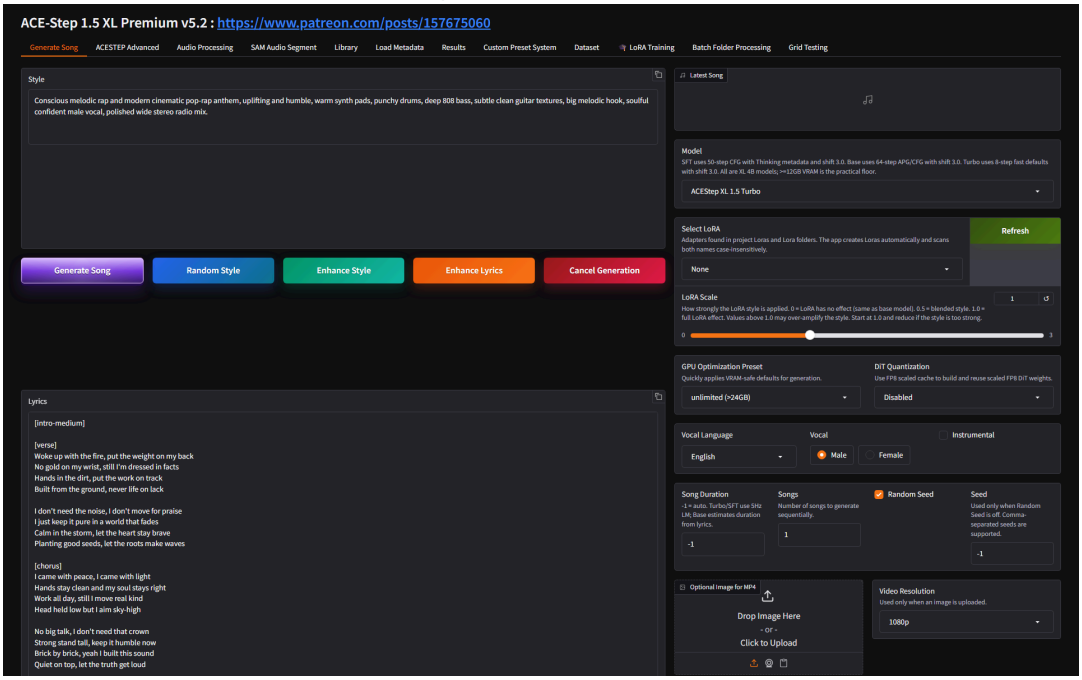
Model availability: Turbo is the quick default. SFT and Base require additional model files. Remix is recommended with SFT in the video; some modes are marked Base-only or unavailable until the matching model is selected.



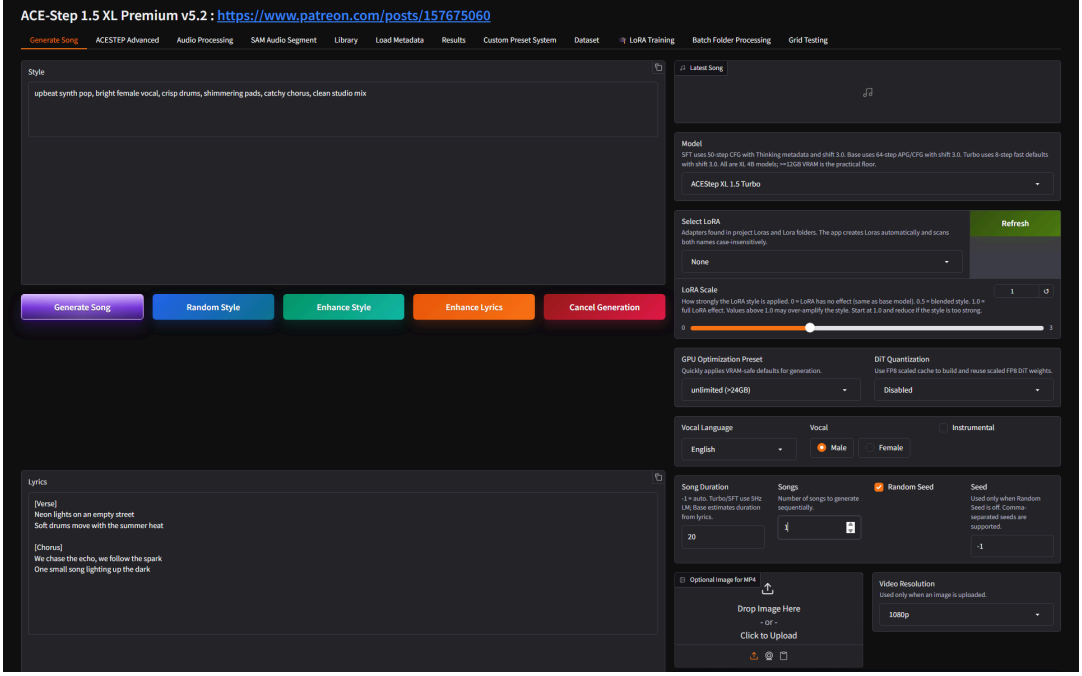
Windows first generation

3. Quick Song Generation

The **Generate Song** tab is the fast path. It exposes the controls most users need: style, lyrics, model, LoRA, GPU preset, quantization, language, vocal type, instrumental toggle, duration, count, seed, optional MP4 image, and video resolution.

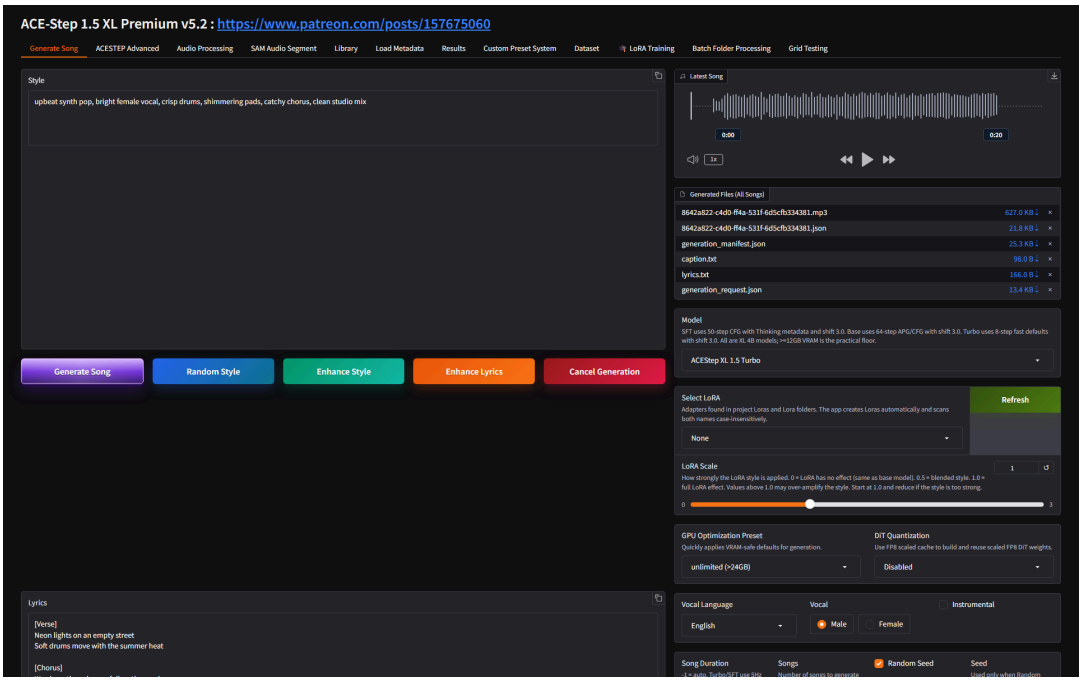


Generate Song overview



Generate Song filled

1. Write a concise Style prompt that describes genre, vocal character, instrumentation, production quality, tempo or mood, and mix target.
2. Write Lyrics with section tags such as `[Verse]` and `[Chorus]`. The included `ACE_Step_Lyric_Generation_Instructions_For_LLMs.txt` file can be given to an LLM to format lyrics or style prompts.
3. Select the Model. Start with **ACE-Step XL 1.5 Turbo** to verify the machine and workflow quickly.
4. Leave GPU Optimization Preset and DiT Quantization at safe defaults unless you are solving VRAM pressure or repeating a known workflow.
5. Set Song Duration and Songs. The demo run used 20 seconds and 1 song.
6. Use Random Seed while exploring. When a promising result appears, uncheck Random Seed and keep the seed so future edits stay comparable.
7. Click **Generate Song** and monitor the Status field plus the terminal window.



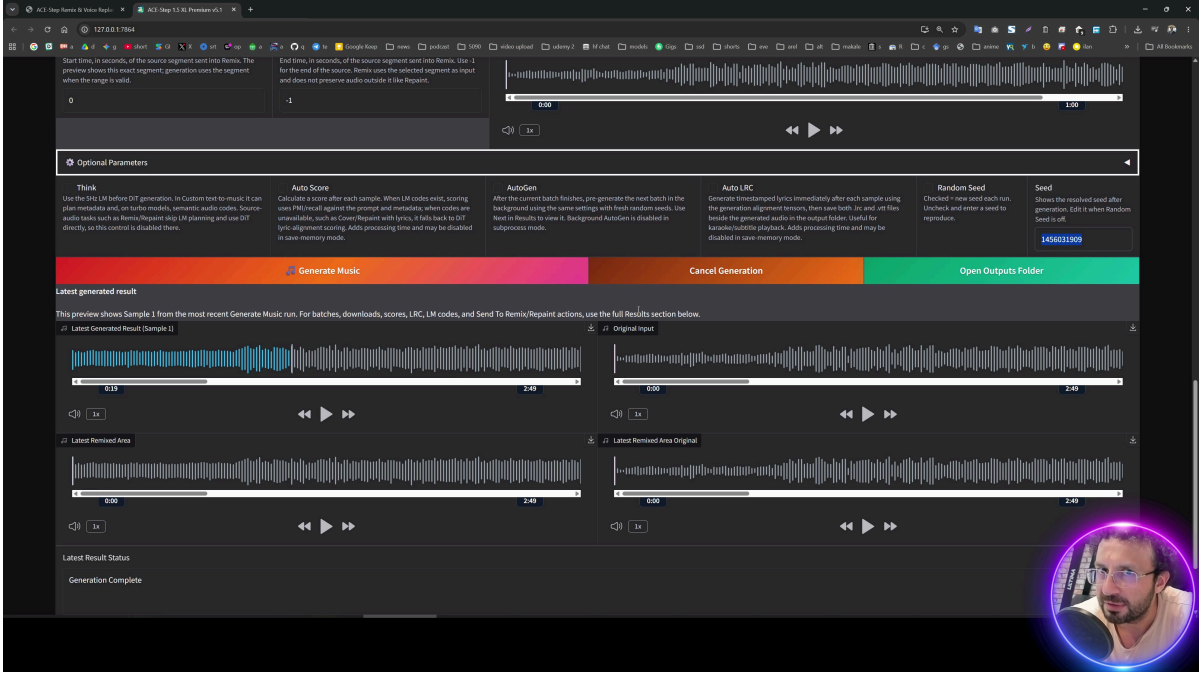
Demo generation result

Useful quick-tab buttons:

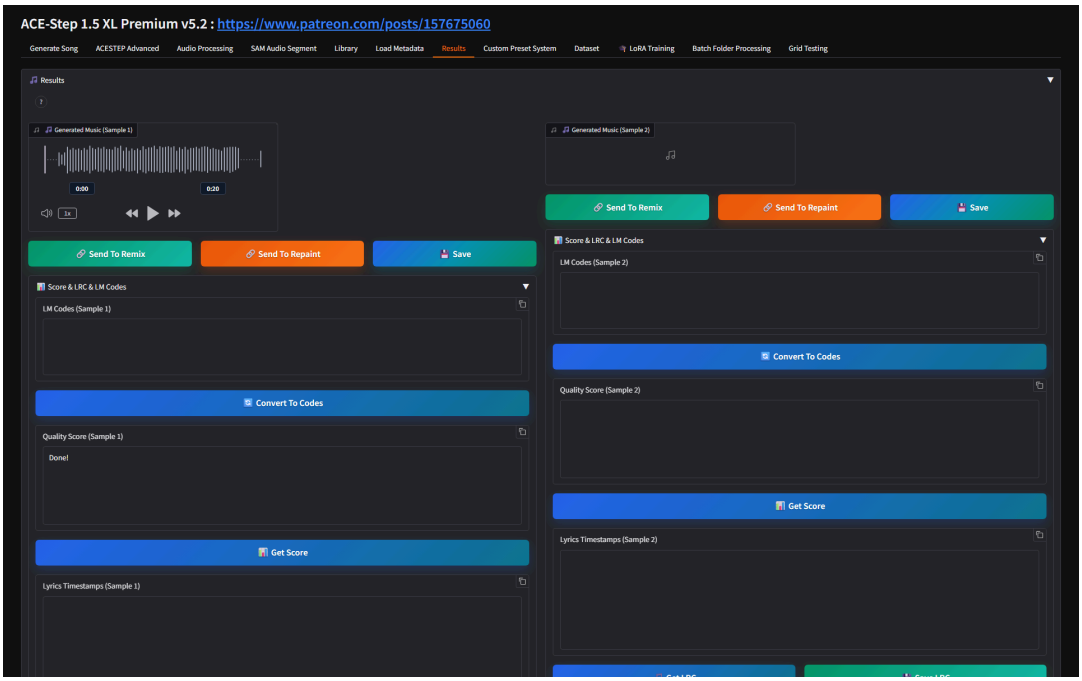
- **Random Style** creates a starting style prompt.
- **Enhance Style** improves the style prompt.
- **Enhance Lyrics** improves lyric structure or phrasing before generation.
- **Cancel Generation** stops a run from the UI.
- **Open Outputs Folder** opens the app's outputs directory where audio, manifests, lyrics, captions, sessions, and metadata are saved.

4. Results, Seeds, And Reuse

The tutorial stresses generating repeatedly until you have a good base result, then locking the seed and making controlled edits. This is especially important for remix and repaint work, where small prompt or range changes can be tested against the same underlying random state.



Seed and remix discussion



Results after generation

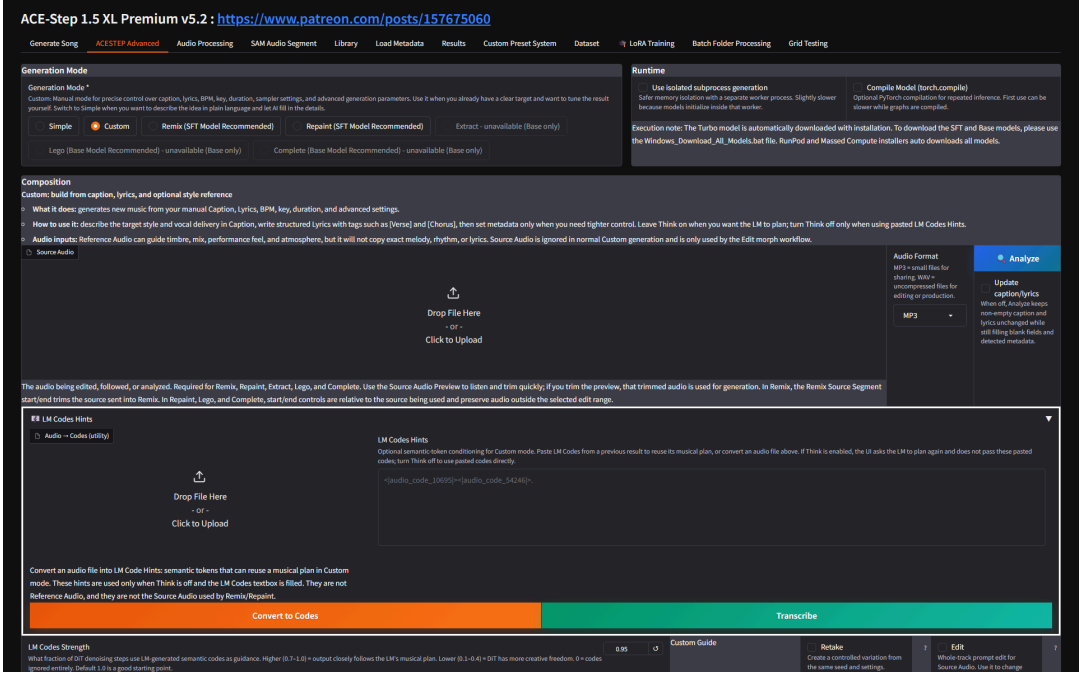
- **Send To Remix** loads the generated song as Source Audio and prepares the advanced Remix workflow.
- **Send To Repaint** loads the generated song and prepares a repaint range workflow.
- **Convert To Codes** reuses the musical plan as LM Codes Hints in compatible Custom workflows.
- **Get Score** and **Get LRC** create quality-score and lyric-timestamp artifacts.
- **Save** and **All Generated Files** expose generated files for download or reuse.

Seed workflow:

1. Keep Random Seed on while searching for a usable base result.
2. When the result is close, copy or keep the seed shown by the UI.
3. Turn Random Seed off.
4. Change one word, one range, or one strength setting at a time.
5. Compare outputs against the locked seed.

5. Advanced Generation Modes

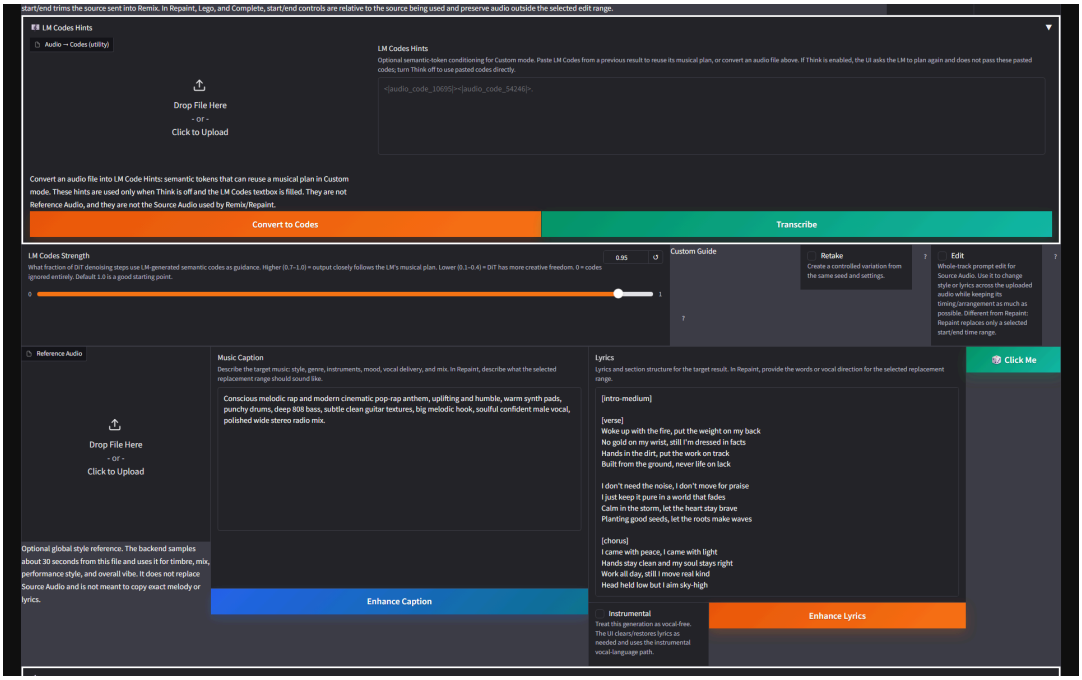
The **ACESTEP Advanced** tab is the full workstation. It exposes generation mode, runtime settings, source/reference audio, LM code utilities, advanced prompts, metadata, sampler settings, output settings, and batch processing.



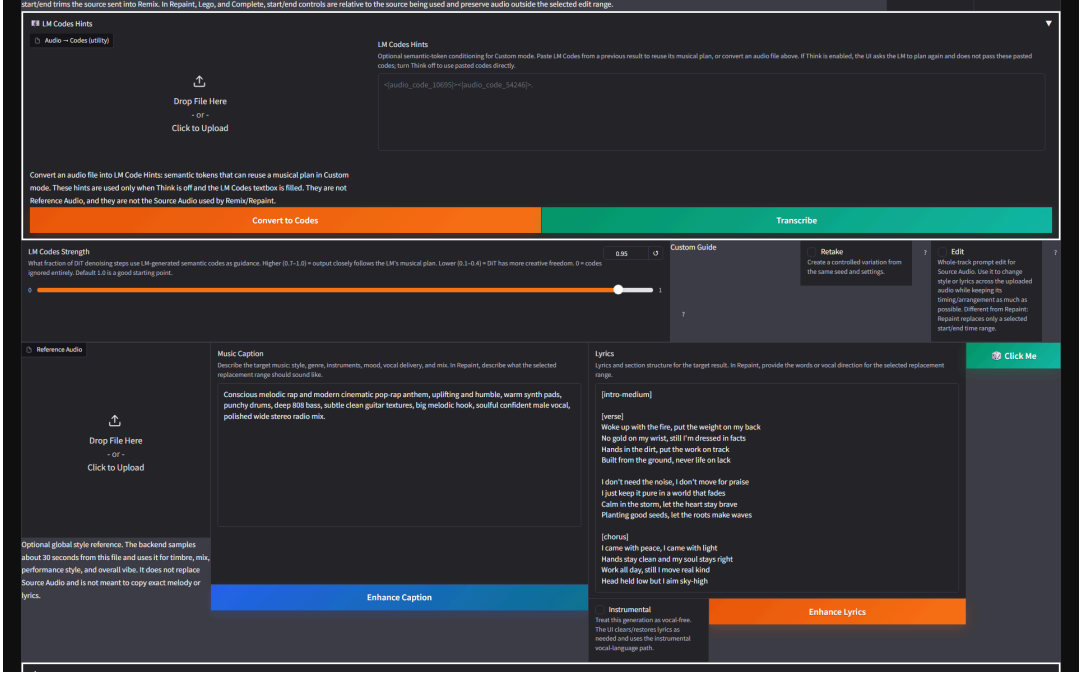
Advanced overview

Generation modes:

- **Simple**: plain-language generation when you want the app to fill many details.
- **Custom**: precise manual control over caption, lyrics, BPM, key, time signature, language, duration, and advanced settings.
- **Remix**: create a new version from source audio. The video recommends SFT for Remix and keeping the same lyrics for best results.
- **Repaint**: replace or modify a selected time range while preserving the rest of the source audio.
- **Extract**: use ACE-Step extraction or stem workflows where the selected model supports it.
- **LEGO**: add a new stem to existing audio, such as adding a guitar stem to a track.
- **Complete**: continue or complete source audio when the selected model supports it.



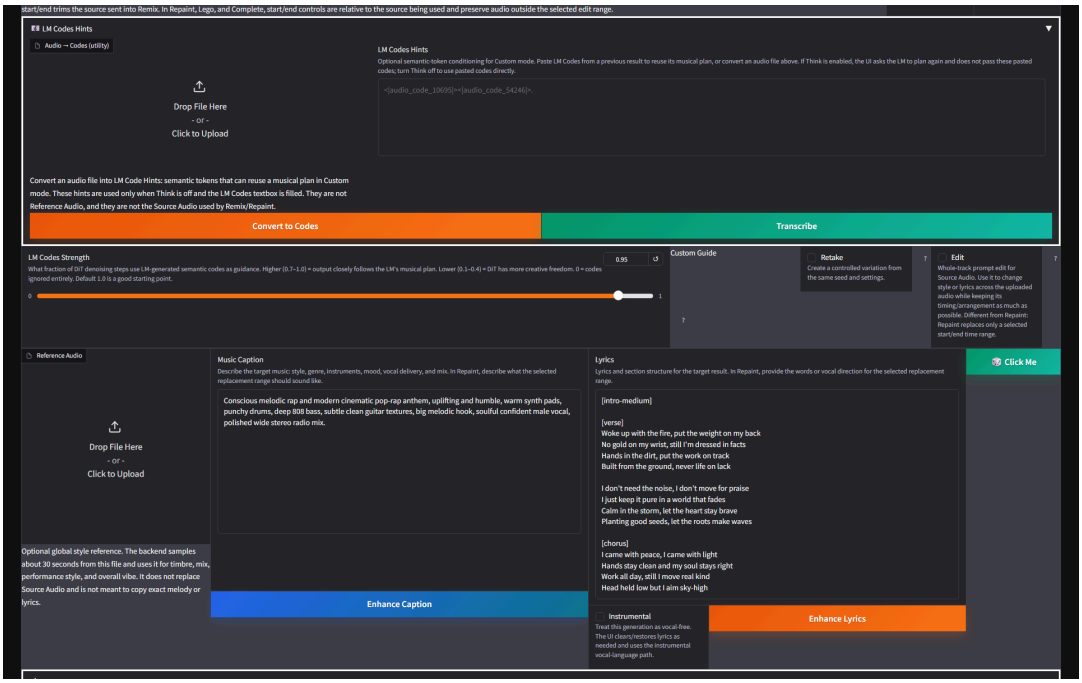
Advanced source audio



Advanced generation controls

Important advanced controls:

- **Source Audio** is required for Remix, Repaint, Extract, LEGO, and Complete.
- **Reference Audio** guides timbre, mix, performance feel, and atmosphere; it is not meant to copy exact melody, rhythm, or lyrics.
- **Analyze** can fill or update caption/lyrics/metadata from source audio.
- **BPM Auto**, **Key Auto**, **TimeSig Auto**, **Language Auto**, and **Duration Auto** let the model infer metadata.
- **Think** enables LM planning. Turn it off only when deliberately using pasted LM Codes Hints.
- **Retake** creates controlled variation from the same seed/settings.
- **Edit** changes the whole uploaded source using source and target prompts.
- **Auto Score**, **AutoGen**, and **Auto LRC** can create score, metadata, and lyric timing outputs during generation.

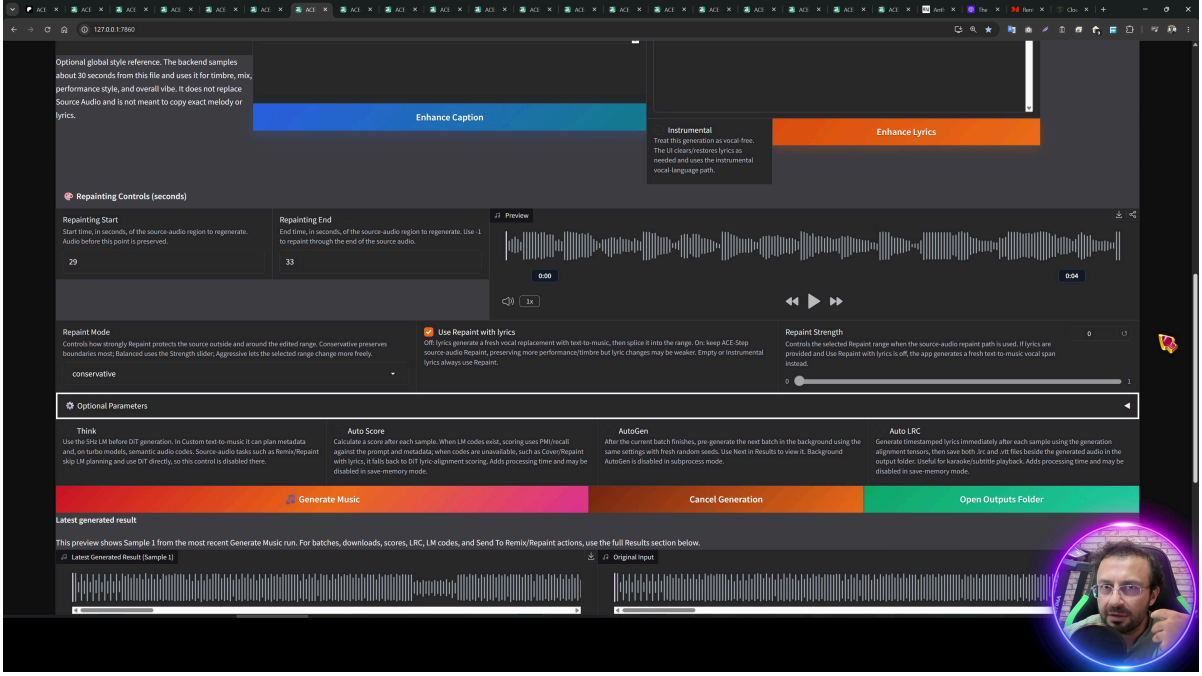


Engine settings

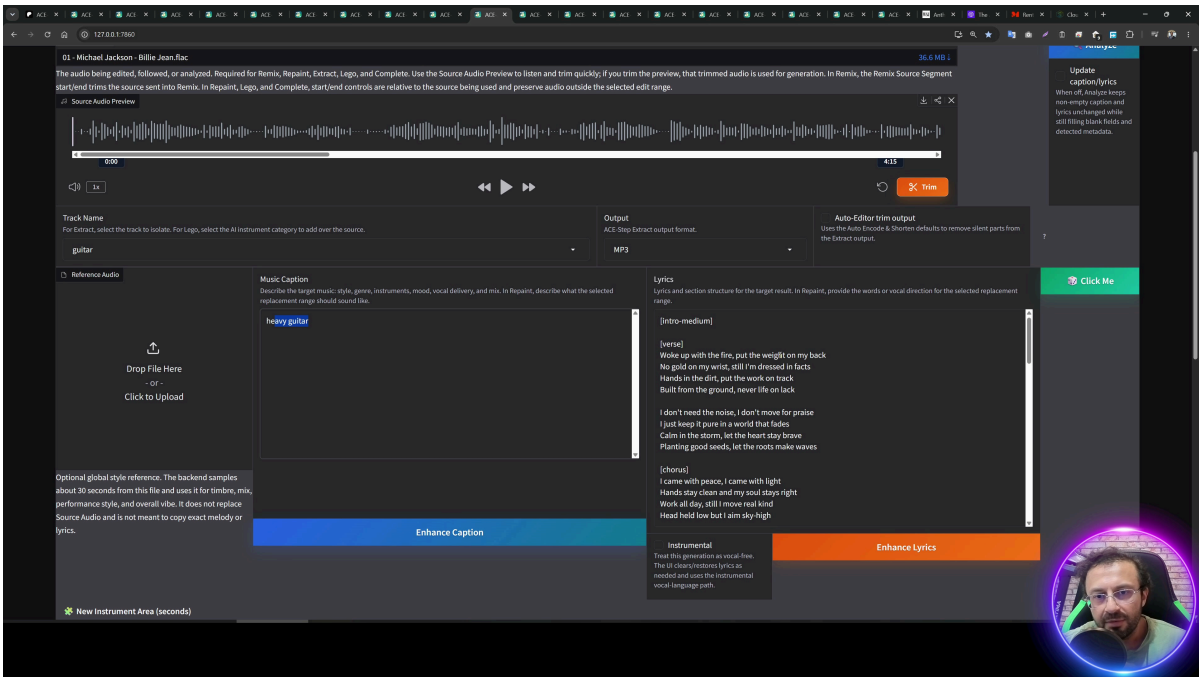
Engine settings include GPU tier, checkpoint file, main model path, device, VAE, 5Hz LM model/backend, Flash Attention, CPU offload, compile, DiT quantization, LoRA path/folder, LoRA scale, inference steps, sampler, DCW, ADG, MP3 bitrate/sample rate, normalization, fades, LM temperature, top-k/top-p, negative prompt, and LM code settings. Leave these at defaults until you have verified a basic generation.

6. Remix, Repaint, Extract, LEGO, And Auto-Editor Features

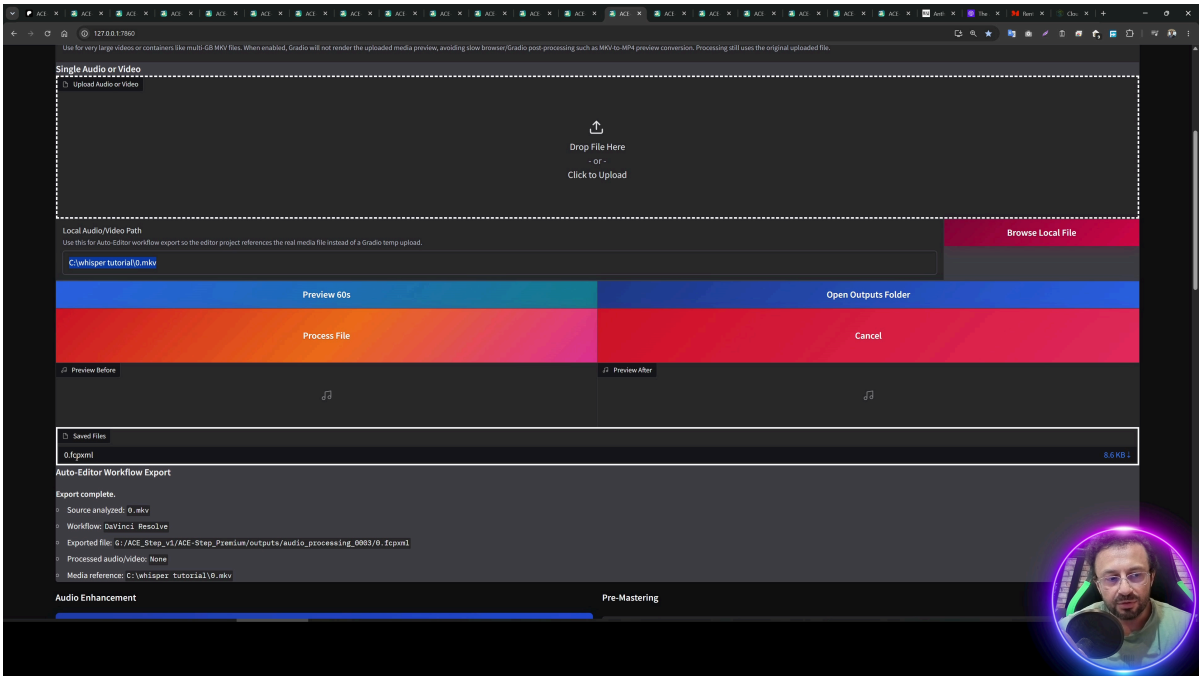
The first part of the video demonstrates feature outcomes before the installation section. These are not separate apps; they are modes and panels inside the same ACE-Step interface.



Remix demo



Extract and LEGO demo

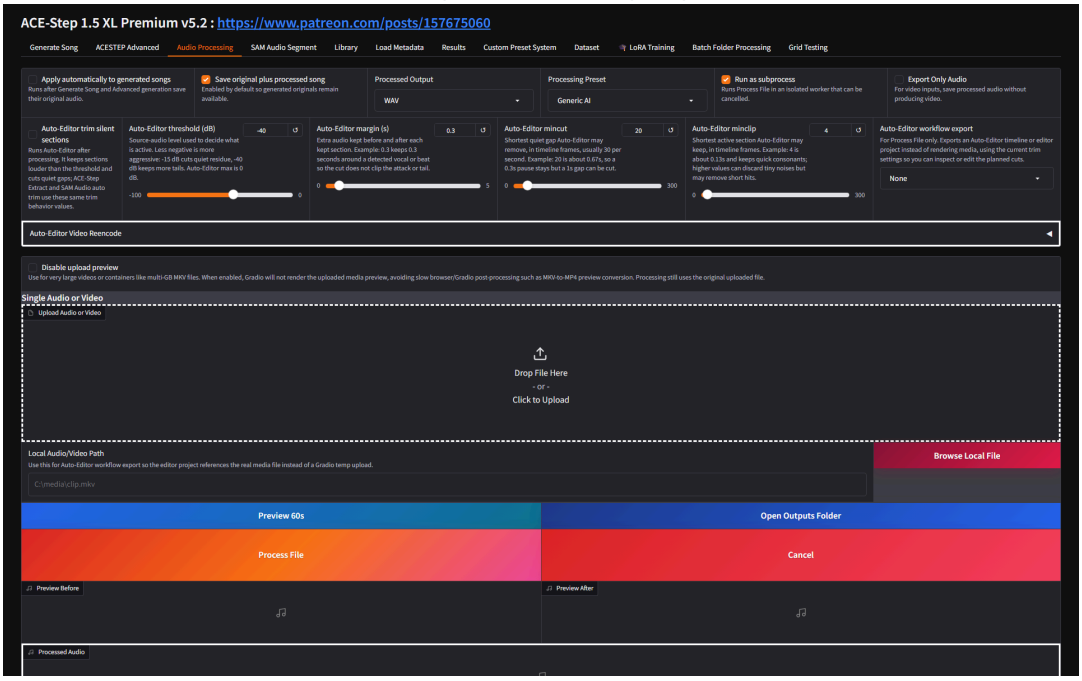


Auto-Editor demo

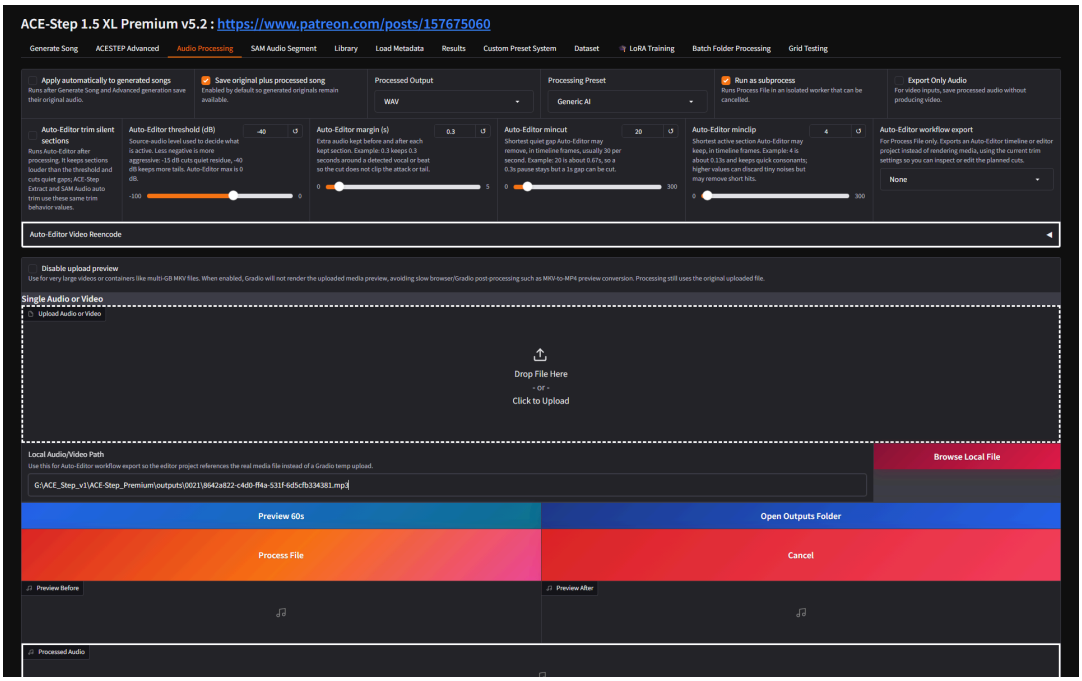
- For **Remix**, upload the source song, keep the same lyrics, use SFT when available, and start with the default remix strength before changing one variable at a time.
- For **Repaint**, set Repainting Start and End carefully, preview the selected range, then choose repaint mode and strength.
- For **Extract**, choose Track Name or Extract All Stems. Batch folder processing can extract from multiple files.
- For **LEGO**, choose or describe the new stem to add and use source audio as the foundation.
- For **Auto-Editor**, set threshold, margin, mincut, and minclip. Workflow export can produce an editor timeline instead of rendering media.

7. Audio Processing

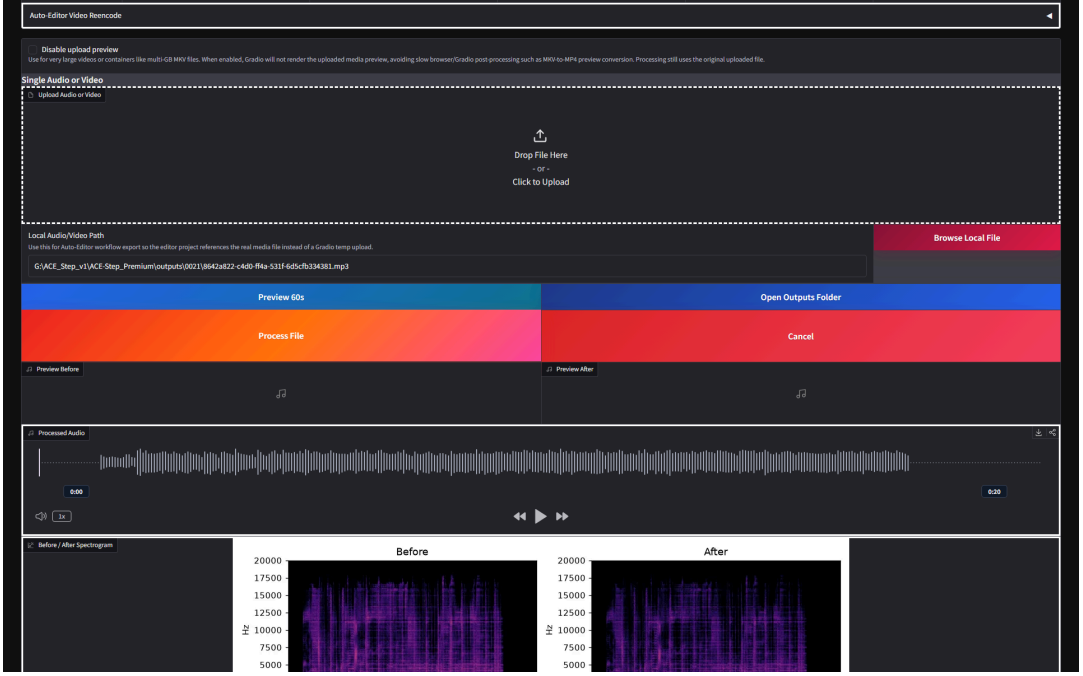
Audio Processing is used on uploaded or local audio/video and can also be applied automatically to generated songs. It includes format output, Auto-Editor trimming, video re-encode controls, audio enhancement stages, pre-mastering stages, DiffPitcher, and batch folder processing.



Audio Processing overview



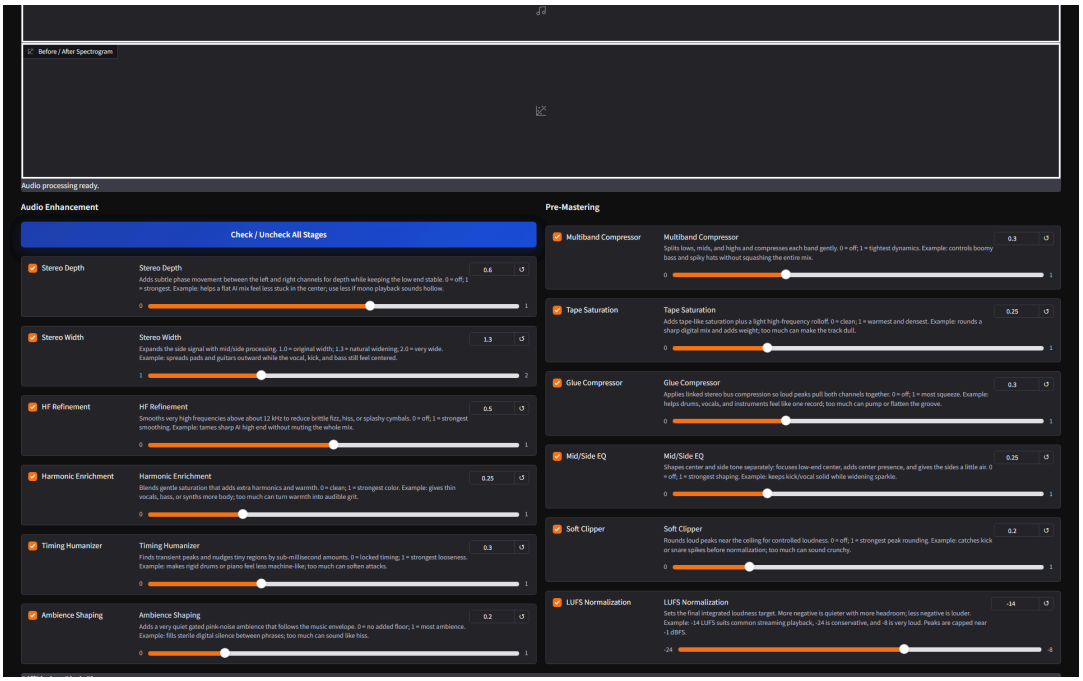
Generated song loaded for processing



Audio Processing result

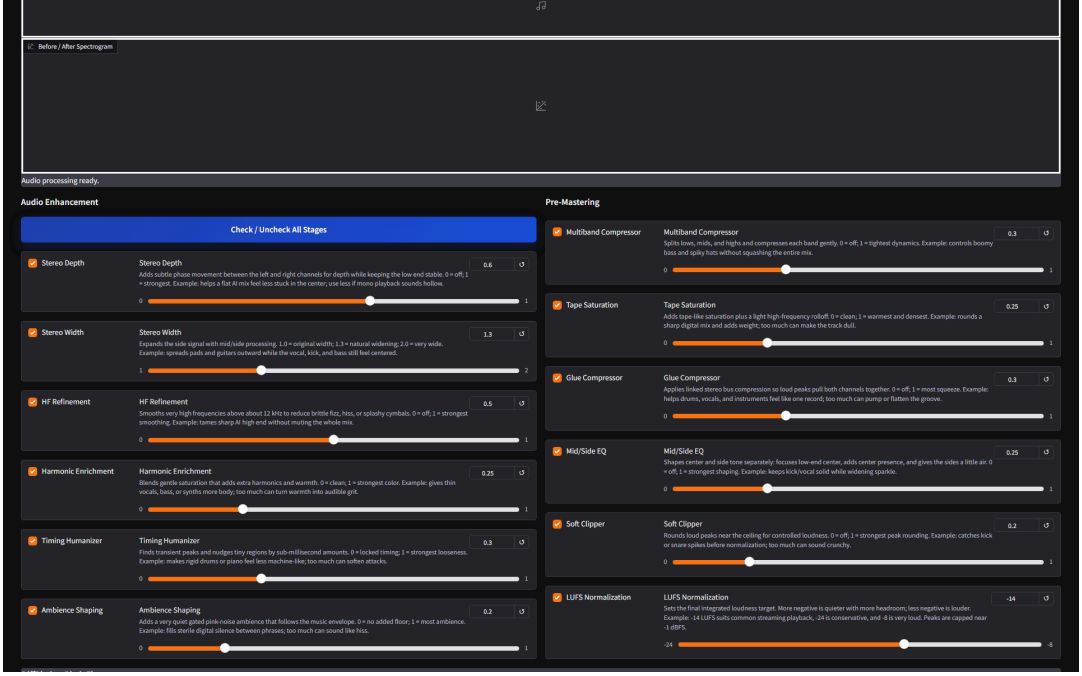
Core Audio Processing controls:

- **Apply automatically to generated songs** runs the processing chain after generation.
- **Save original plus processed song** keeps an untouched copy beside the processed copy.
- **Processed Output** selects WAV/MP3 or another output format.
- **Processing Preset** sets a preset chain before you tune individual stages.
- **Run as subprocess** isolates processing so cancellation and memory cleanup are safer.
- **Export Only Audio** extracts processed audio from video inputs.
- **Auto-Editor trim silent sections** removes quiet/silent segments using threshold, margin, mincut, and minclip.
- **Auto-Editor workflow export** exports an editing timeline/workflow instead of only rendering media.
- **Disable upload preview** helps with very large MKV or multi-GB media.



Audio Enhancement and Pre-Mastering

Audio Enhancement includes Stereo Depth, Stereo Width, HF Refinement, Harmonic Enrichment, Timing Humanizer, and Ambience Shaping. Pre-Mastering includes Multiband Compressor, Tape Saturation, Glue Compressor, Mid/Side EQ, Soft Clipper, and LUFS Normalization.

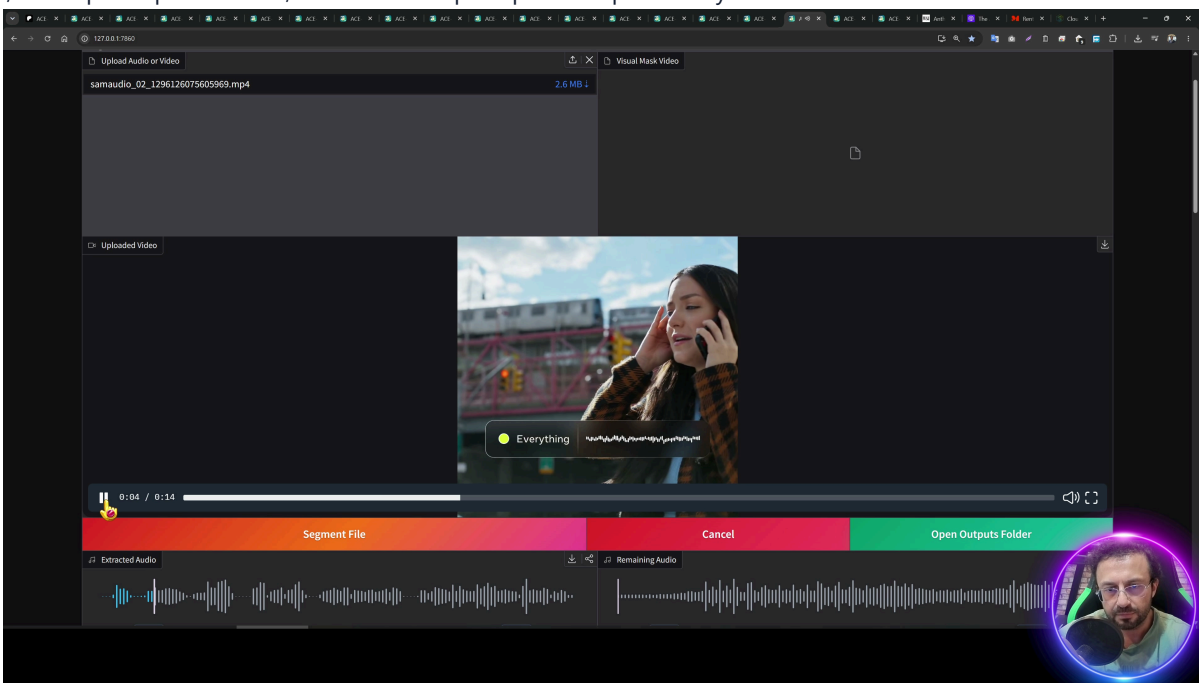


DiffPitcher controls

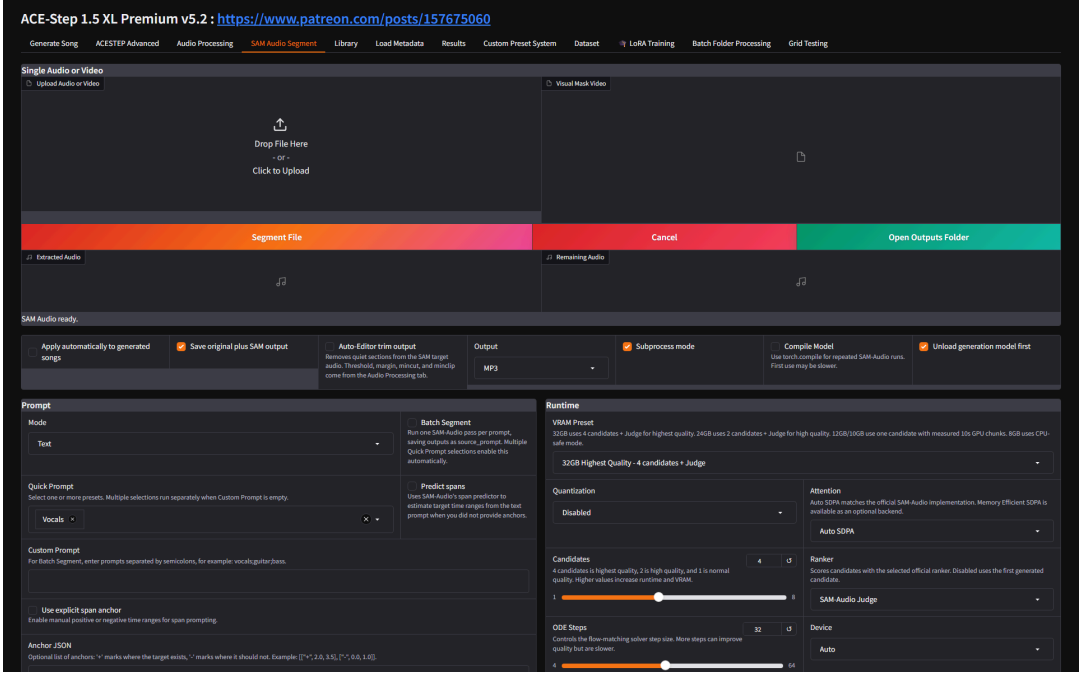
DiffPitcher is for isolated vocals that sing the wrong notes. Use a guide vocal or MIDI score for the same phrase. The tutorial text in the UI warns that this is not for copying another singer or another song.

8. SAM Audio Segment

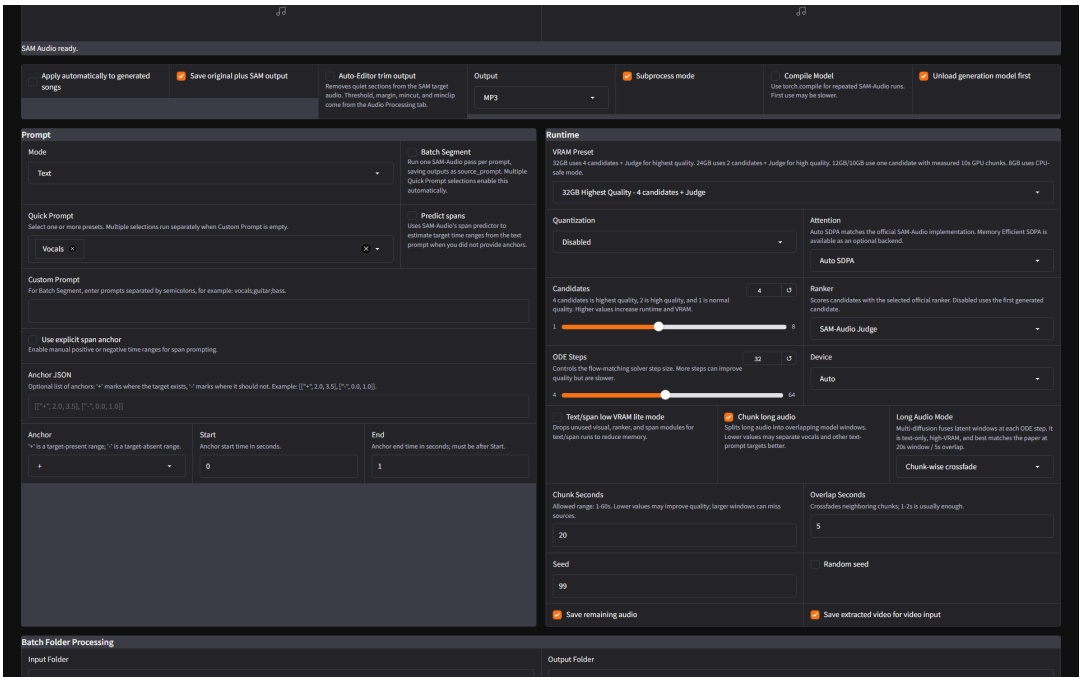
SAM Audio Segment is a heavier but more flexible segmentation system. It can extract target audio from a prompt, save the residual/remaining audio, process video inputs, use explicit span anchors, and run batch prompt lists separated by semicolons.



SAM Audio source-video demo



SAM Audio overview

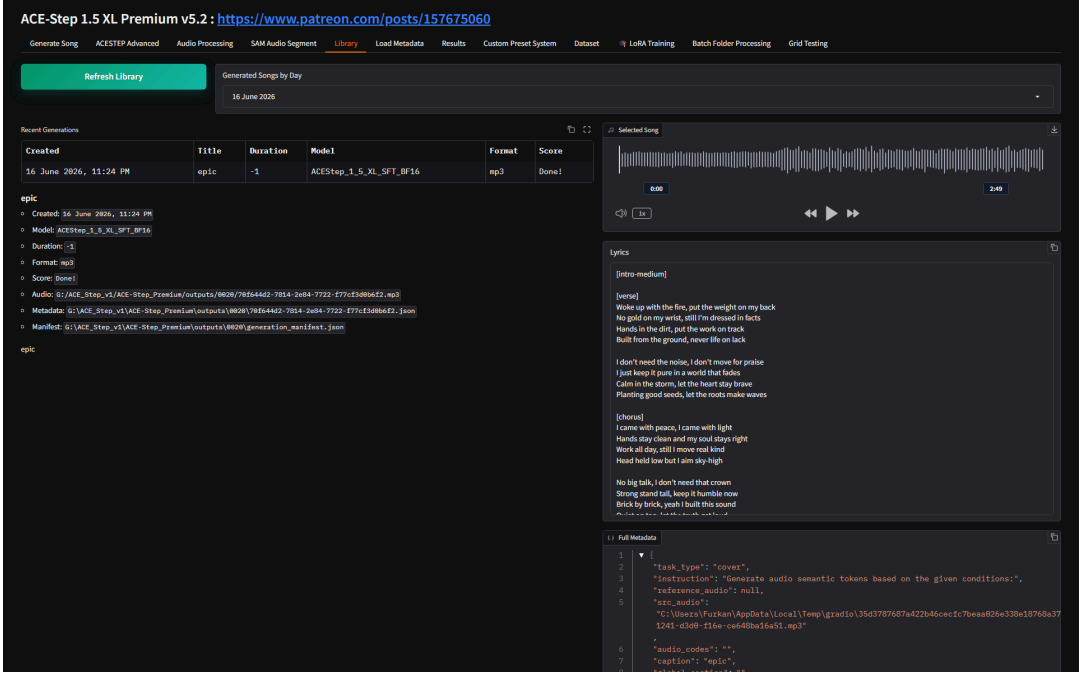


SAM prompt runtime controls

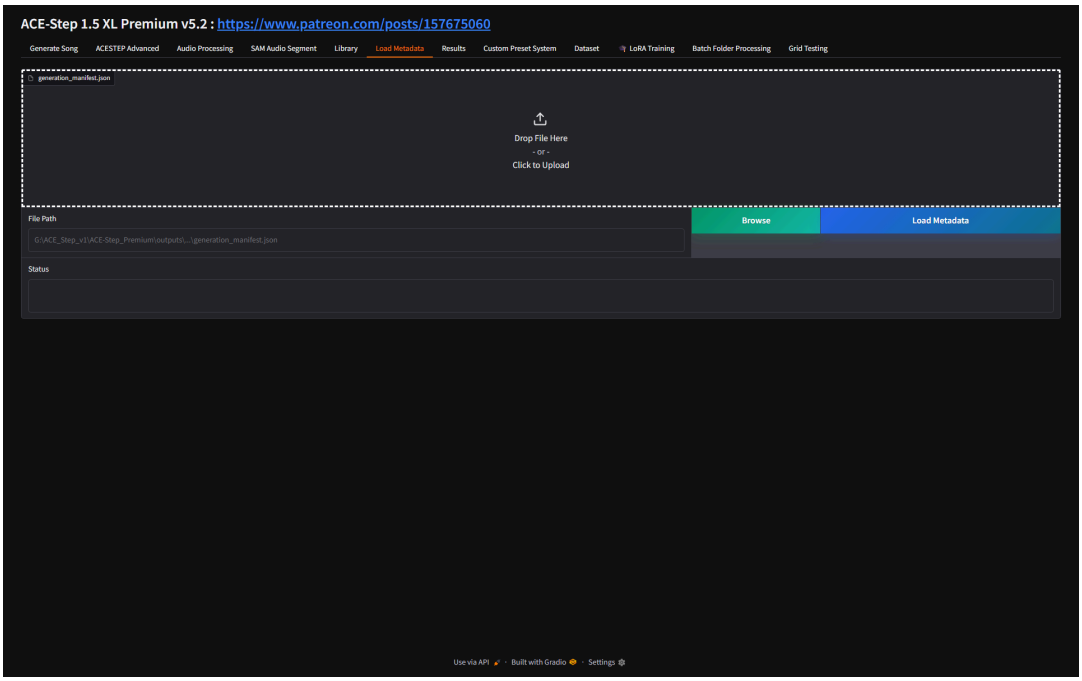
1. Upload an audio or video file. Optionally upload a visual mask video for video-guided workflows.
2. Choose Mode and Quick Prompt, or type a Custom Prompt such as vocals, guitar, bass, drums, applause, or another target.
3. Enable Batch Segment when you want several prompts in one run; separate prompts with semicolons.
4. Use Predict spans when you want SAM Audio to estimate target time ranges from text.
5. Use explicit span anchor only when you can provide positive/negative time anchors as JSON.
6. Choose a VRAM preset and candidate count that match the GPU. Higher candidate counts can improve quality but cost runtime and VRAM.
7. Enable Save remaining audio when you need both the extracted target and the residual track.

9. Library, Metadata, Presets, Dataset, And Training Pages

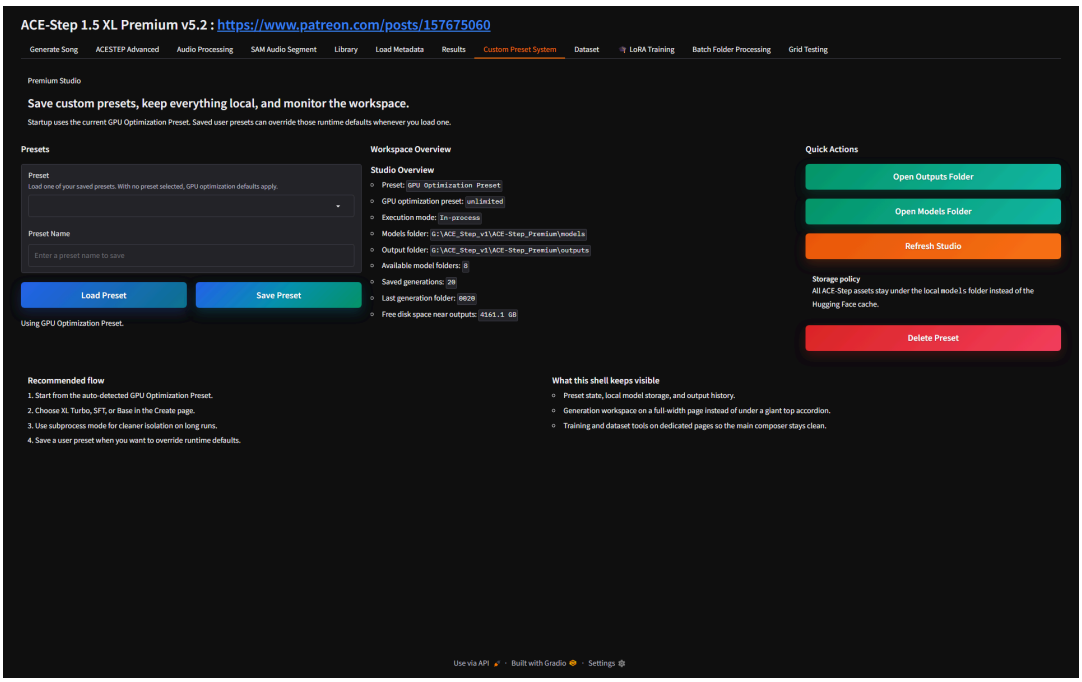
The remaining app tabs are operational pages. They help you find previous generations, restore metadata, manage presets, inspect datasets, and train adapters.



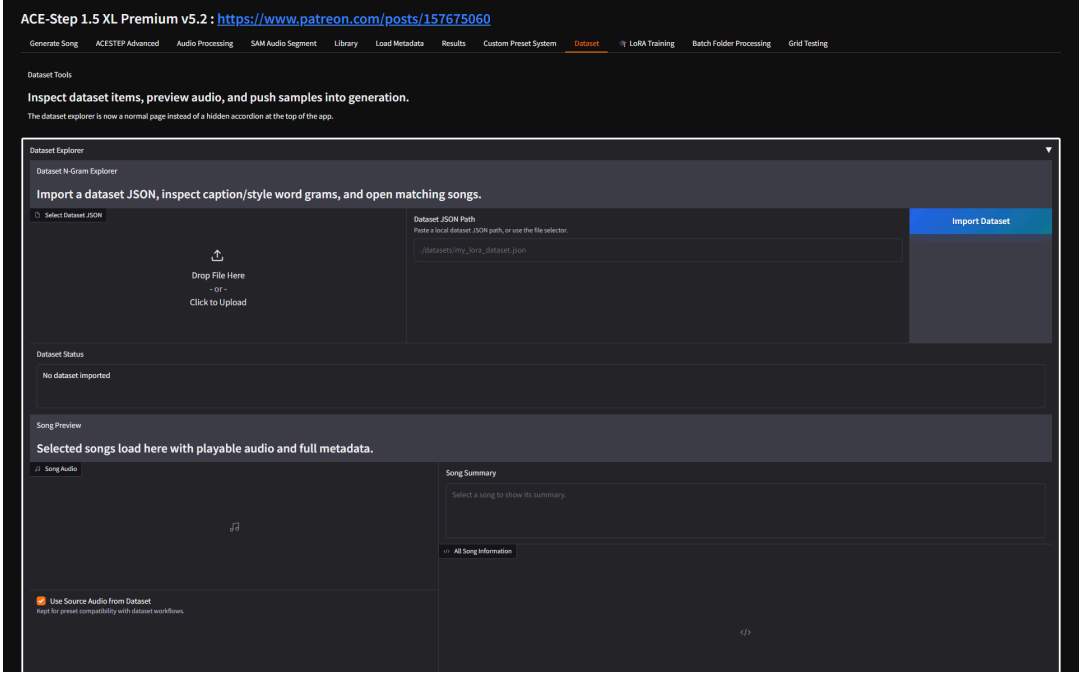
Library



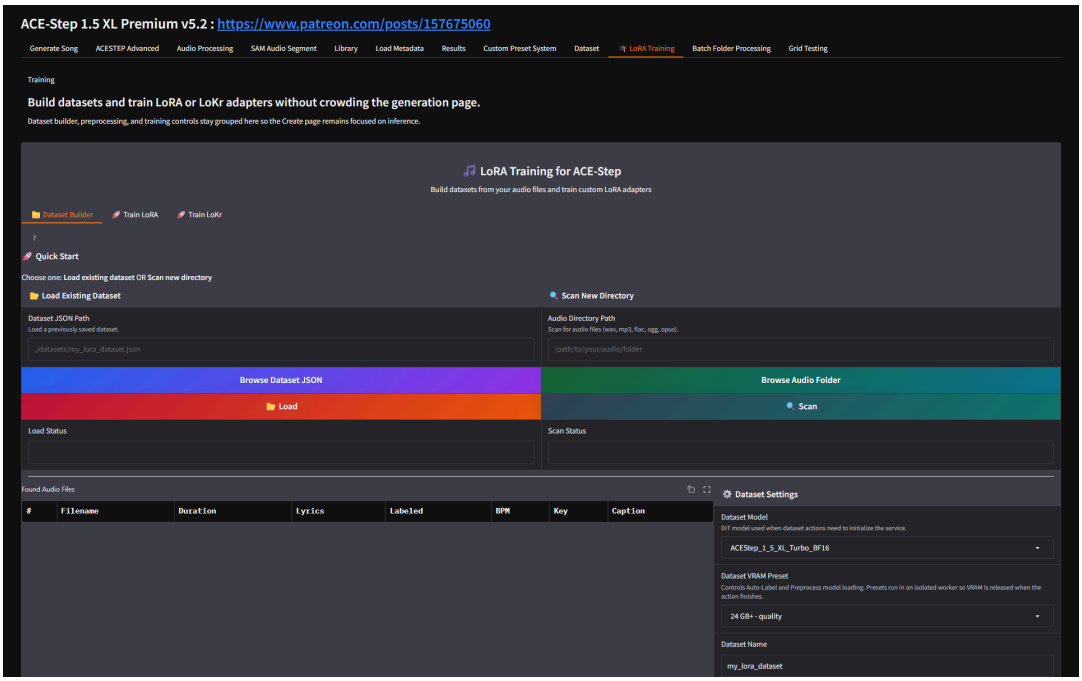
Load Metadata



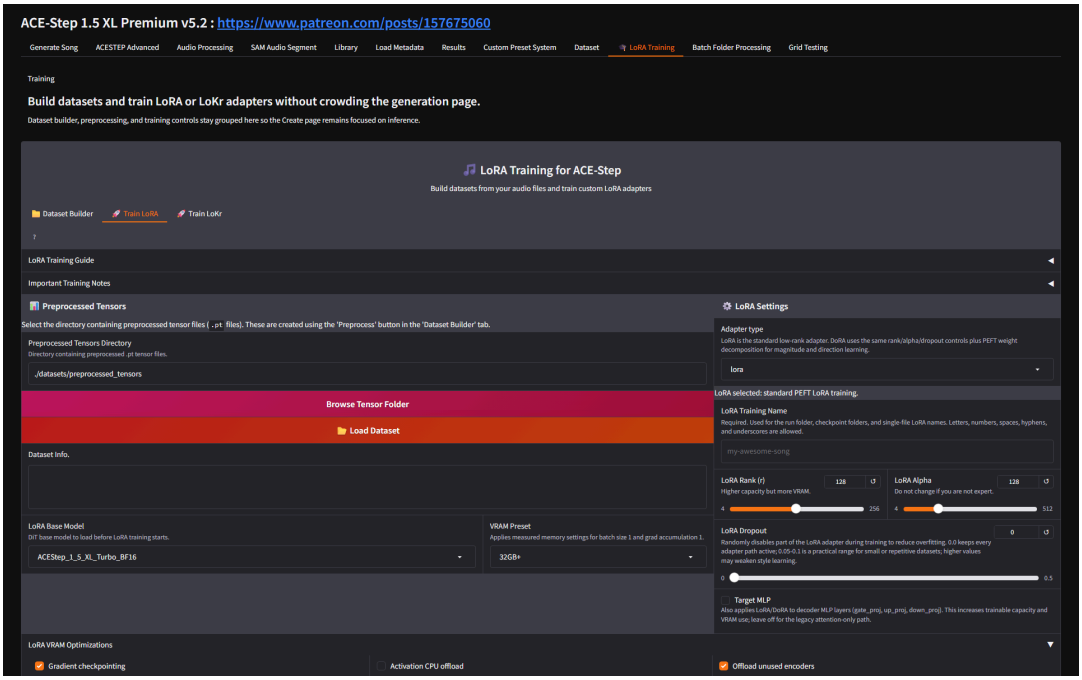
Custom Preset System



Dataset browser



LoRA Dataset Builder

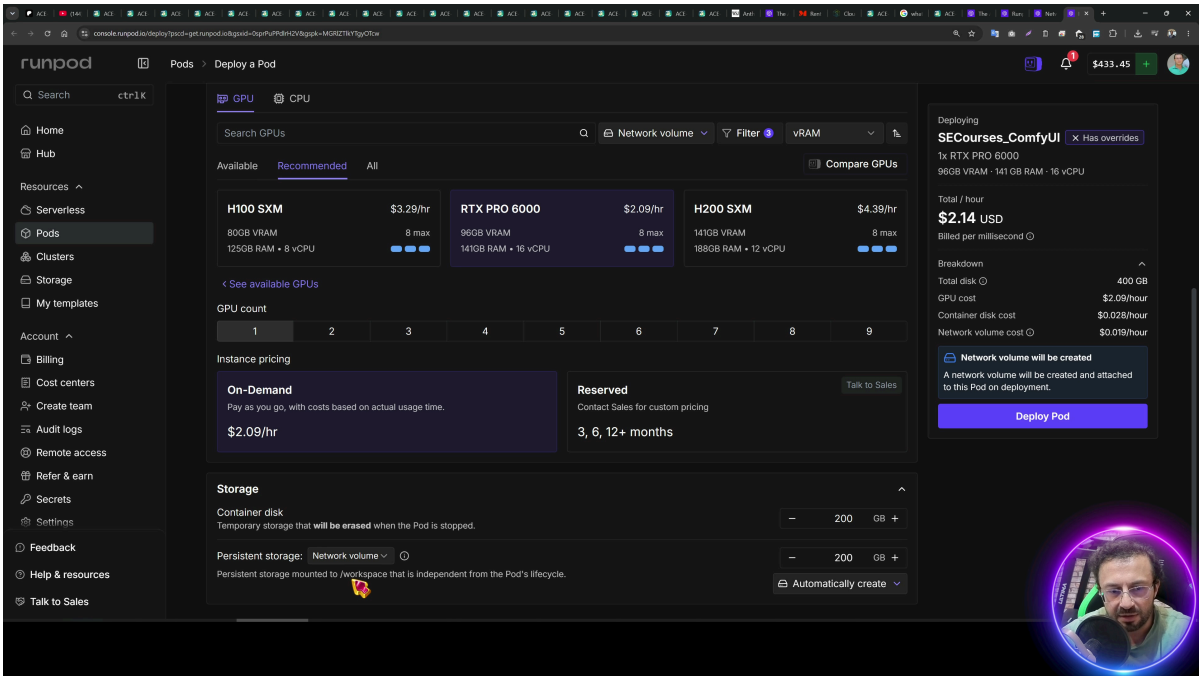


Train LoRA

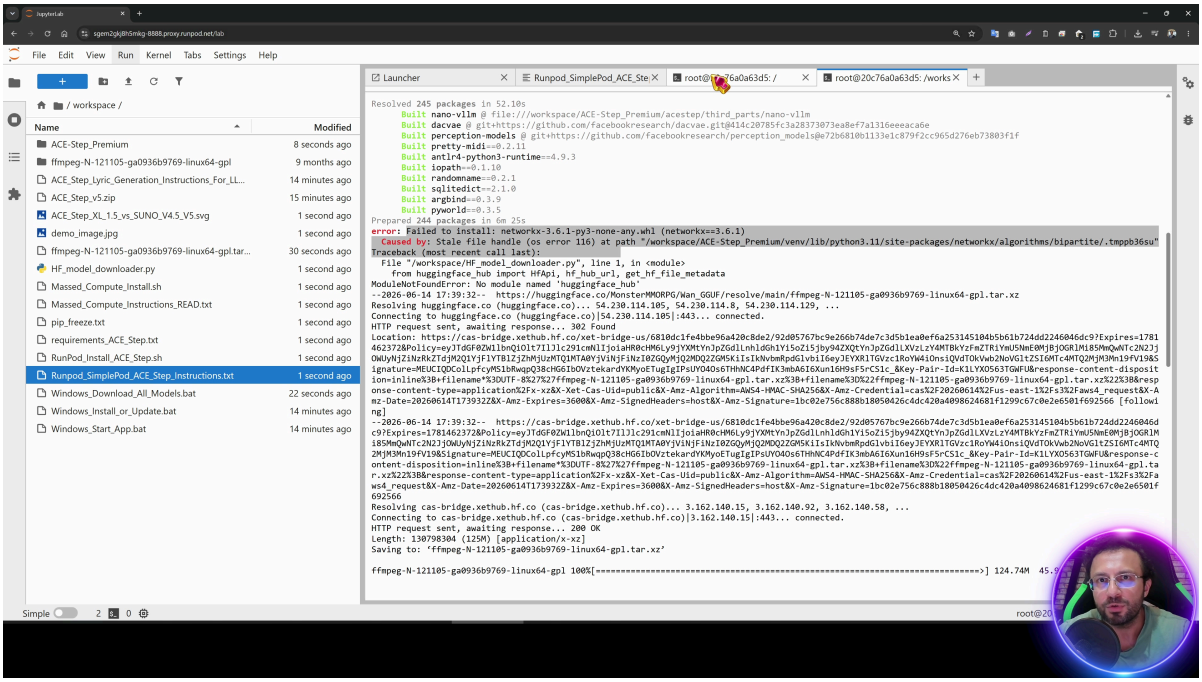
- Use **Library** when you want to find a past song by day and inspect its lyrics/metadata.
- Use **Load Metadata** when you have a `generation_manifest.json` and want to restore a generation into the UI.
- Use **Custom Preset System** to persist frequently used model, GPU, LoRA, audio, and generation defaults across sessions.
- Use **Dataset Builder** to scan audio, auto-label captions/lyrics/BPM/key/time signature, review samples, and save a dataset JSON.
- Use **Preprocess** before training; it creates tensor files for faster LoRA or LoKr training.
- Use **Train LoRA** or **Train LoKr** only after preparing a clean dataset. The video states LoRA training is intended for a separate deeper tutorial.

10. RunPod Deployment

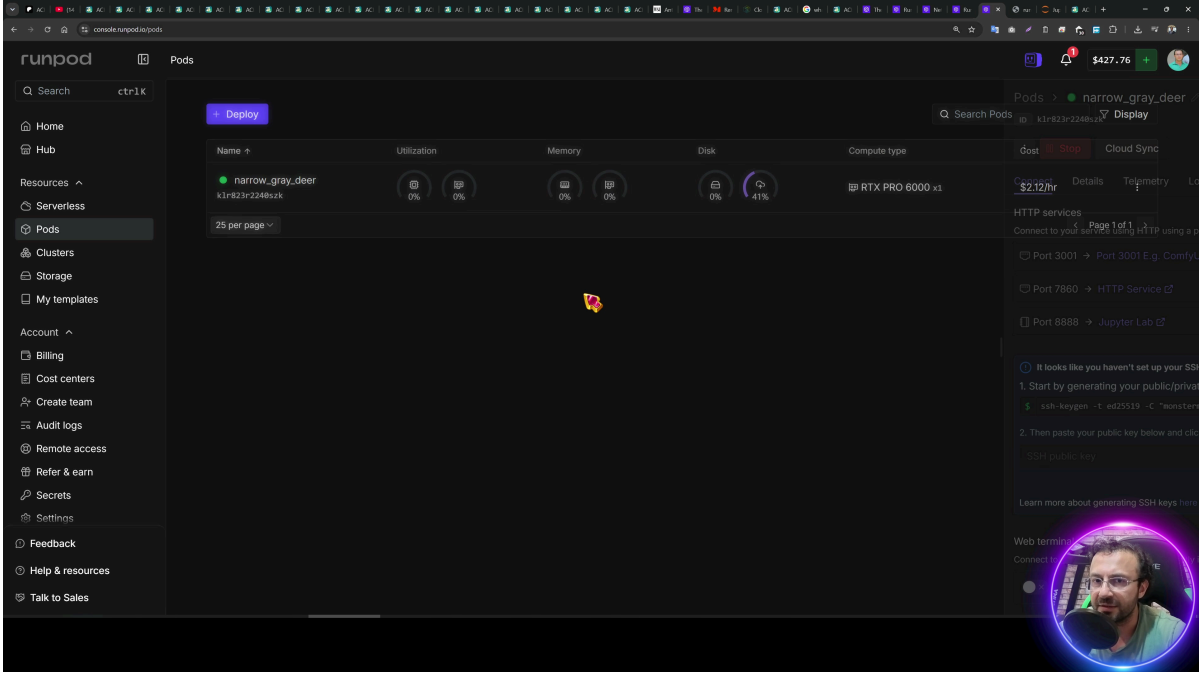
The RunPod chapter focuses on persistent network storage, GPU/region selection, unreliable installs, Gradio live URLs, nvitop monitoring, output downloads, and safe termination.



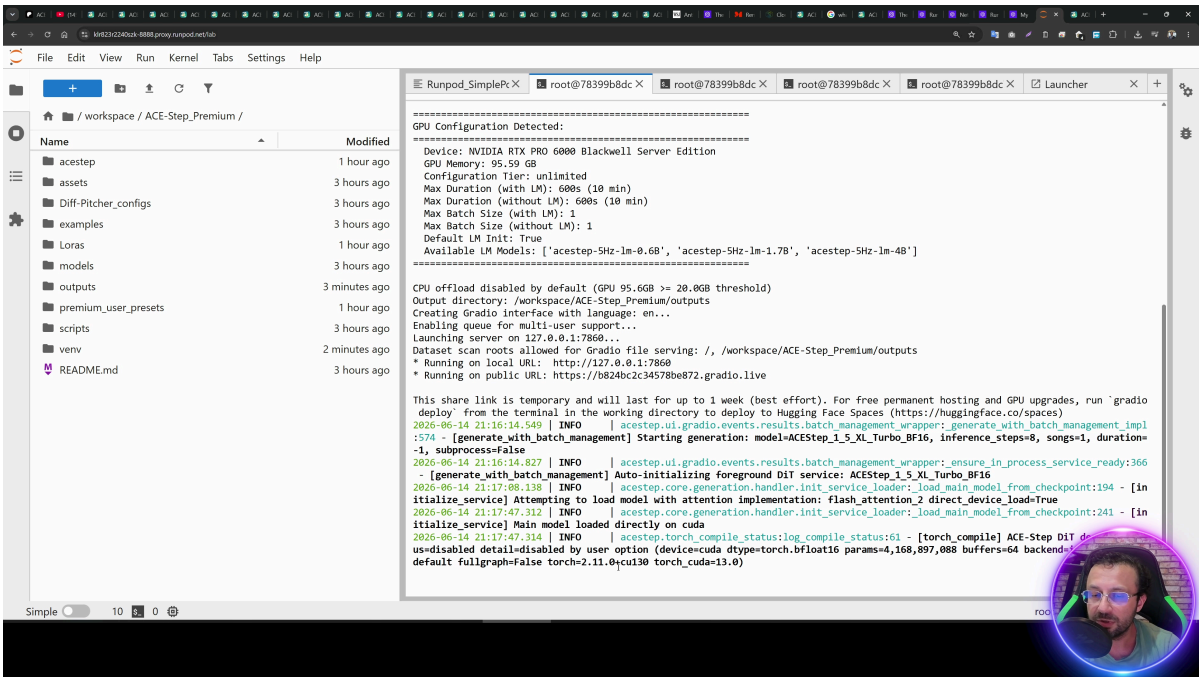
RunPod storage



RunPod install



RunPod Gradio services

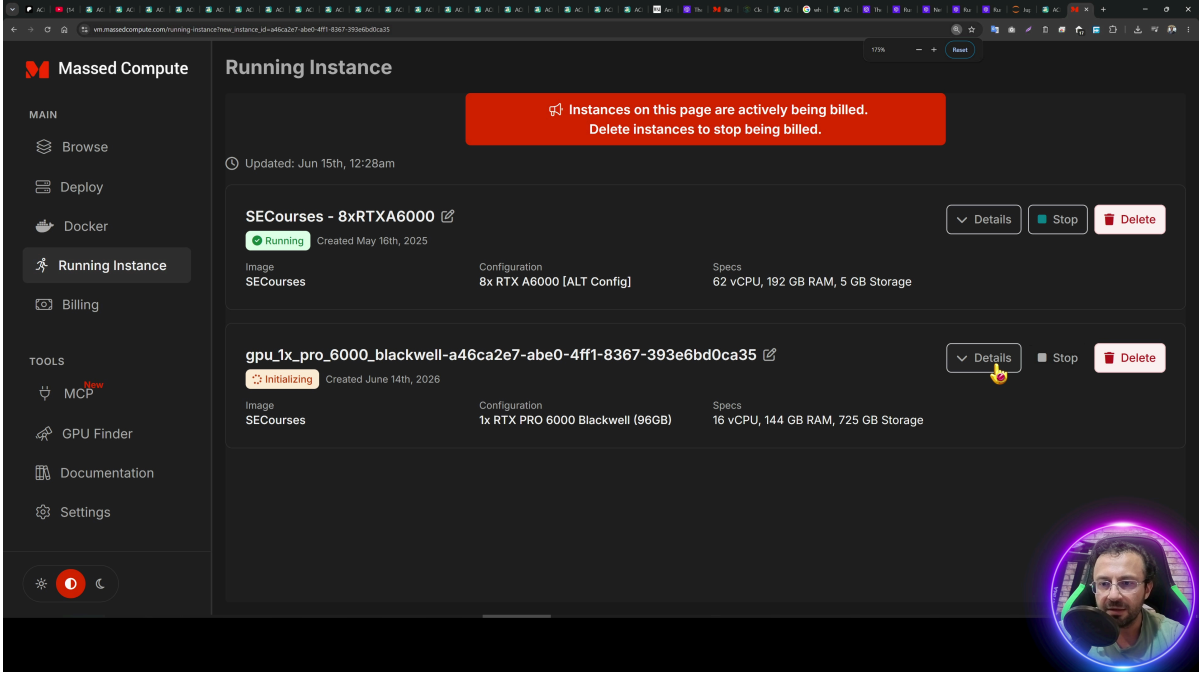


RunPod monitoring

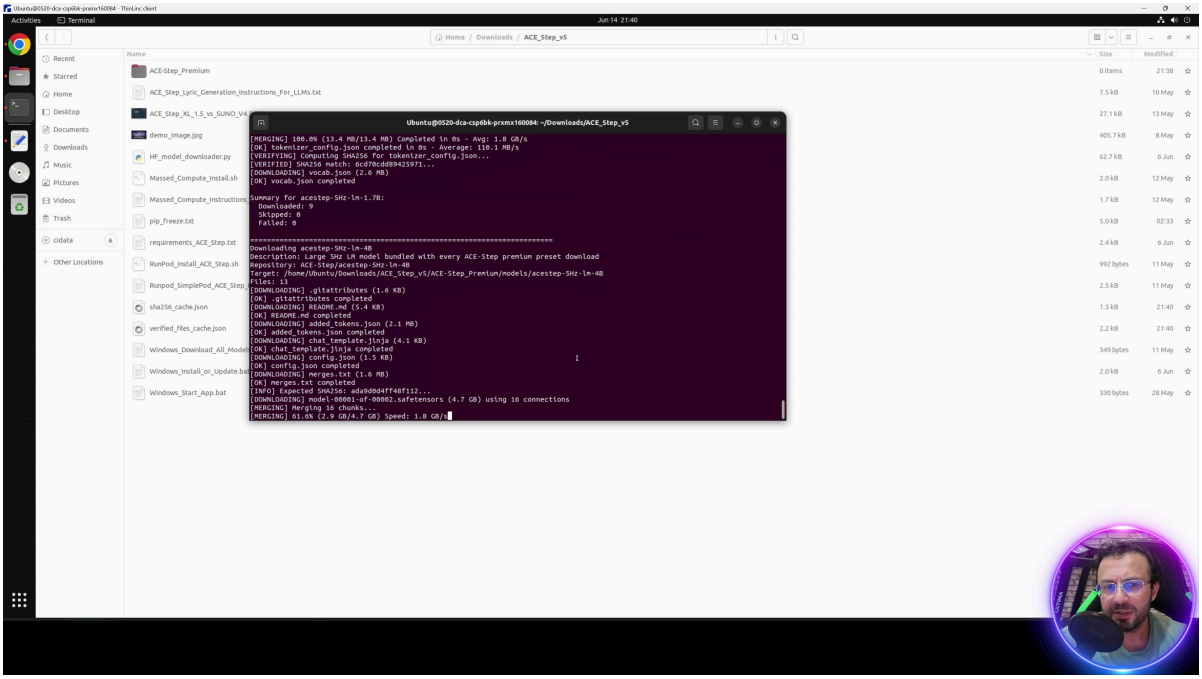
1. Create persistent network storage in the same region as the GPU you intend to rent.
2. Deploy the pod/template with the storage mounted. Choose a GPU with enough VRAM for the selected model and quality target.
3. Run the installer. If RunPod throws an OS/server error, run the installer again; it should resume from completed work.
4. If installation stalls from excessive parallelism, delete the virtual environment, lower installer thread count as shown in the video, and rerun.
5. Start the app and prefer the Gradio live link when the RunPod proxy is unreliable. If port `7860` does not open, try the port shown by the terminal, sometimes `7861`.
6. Use `nvitop` to monitor GPU memory and load. First model load can be slow on RunPod storage; later generations are faster.
7. Download outputs from JupyterLab by right-clicking the outputs folder and downloading it as an archive.
8. Stop or terminate the pod deliberately. Delete storage too if you no longer want monthly storage charges.

11. Massed Compute Deployment

The Massed Compute chapter is similar to the Linux/cloud workflow, but the tutorial emphasizes faster disk performance and lower friction compared with RunPod. The tradeoff called out in the video is the lack of the same persistent network storage flow.



Massed Compute GPU selection

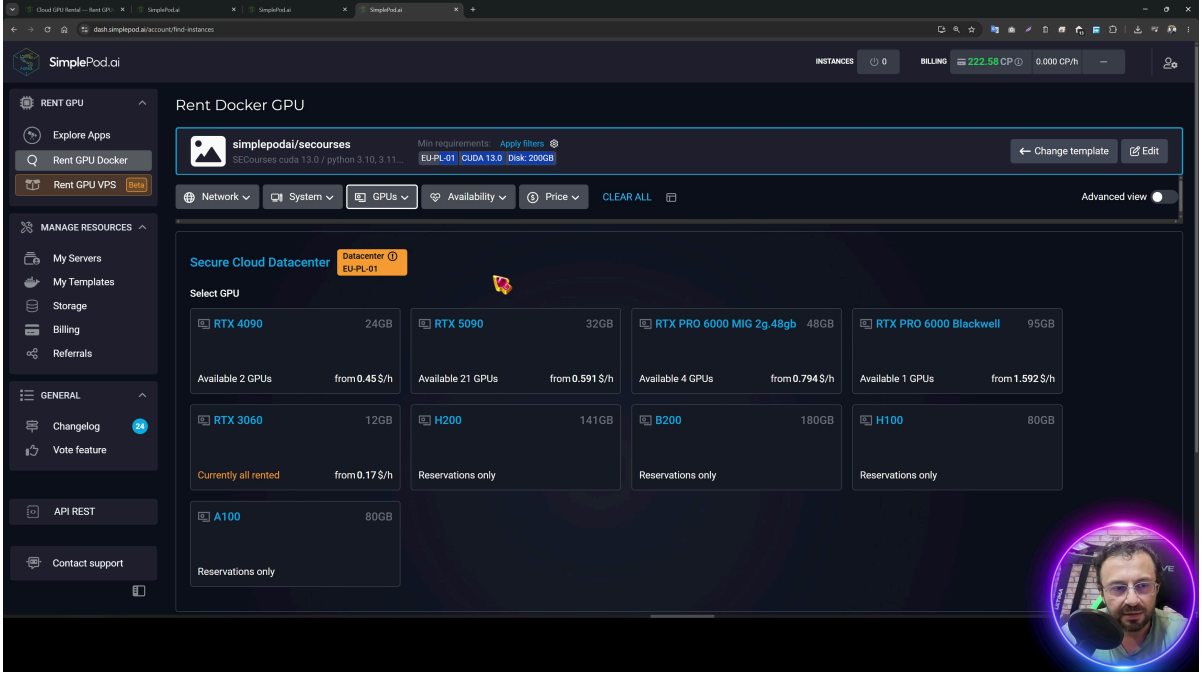


Massed Compute install

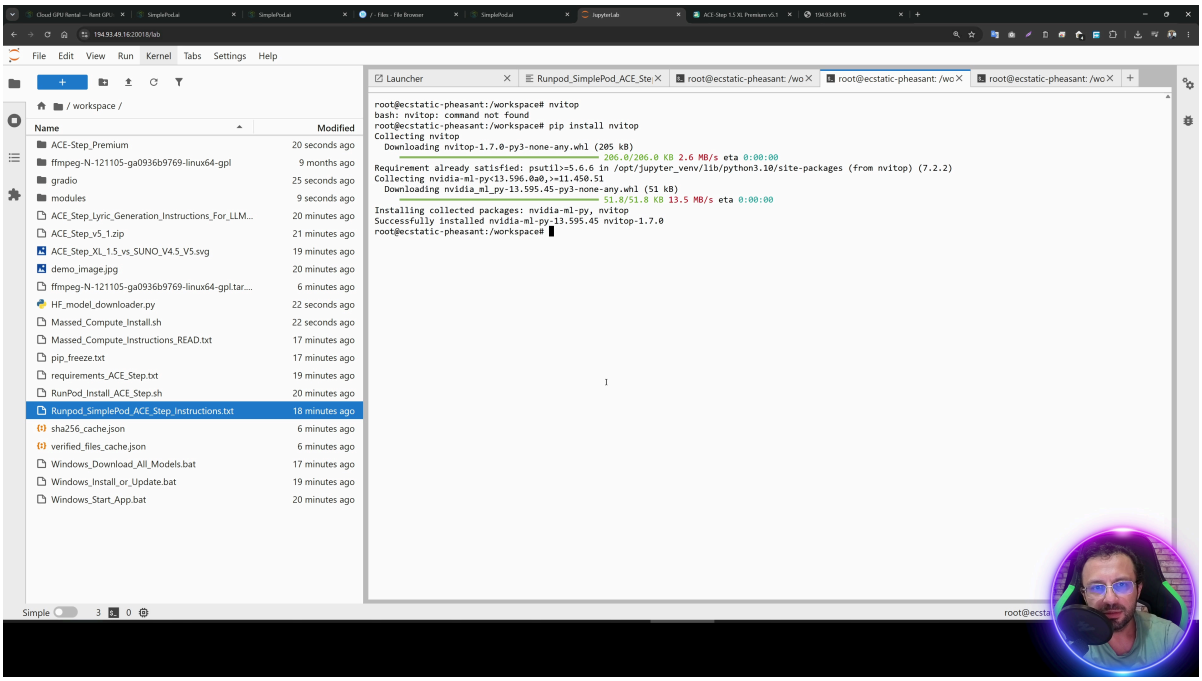
1. Choose the creator category and the SECourses image when following the video workflow.
2. Select a GPU appropriate for ACE-Step XL 1.5. The tutorial mentions RTX Pro 6000 and RTX 5090 class GPUs.
3. Upload the ACE-Step ZIP to Downloads, extract it, open `Massed_Compute_Instructions_READ.txt`, and copy the install command.
4. Open a terminal inside the extracted ACE-Step folder and run the command from that location.
5. Start ACE-Step and use the Gradio live URL. If Gradio live shows a transient error, refresh the page.
6. Back up large outputs or model/data folders to Hugging Face, Google Drive, OneDrive, or another storage service if you need to recreate the machine later.

12. SimplePod Deployment

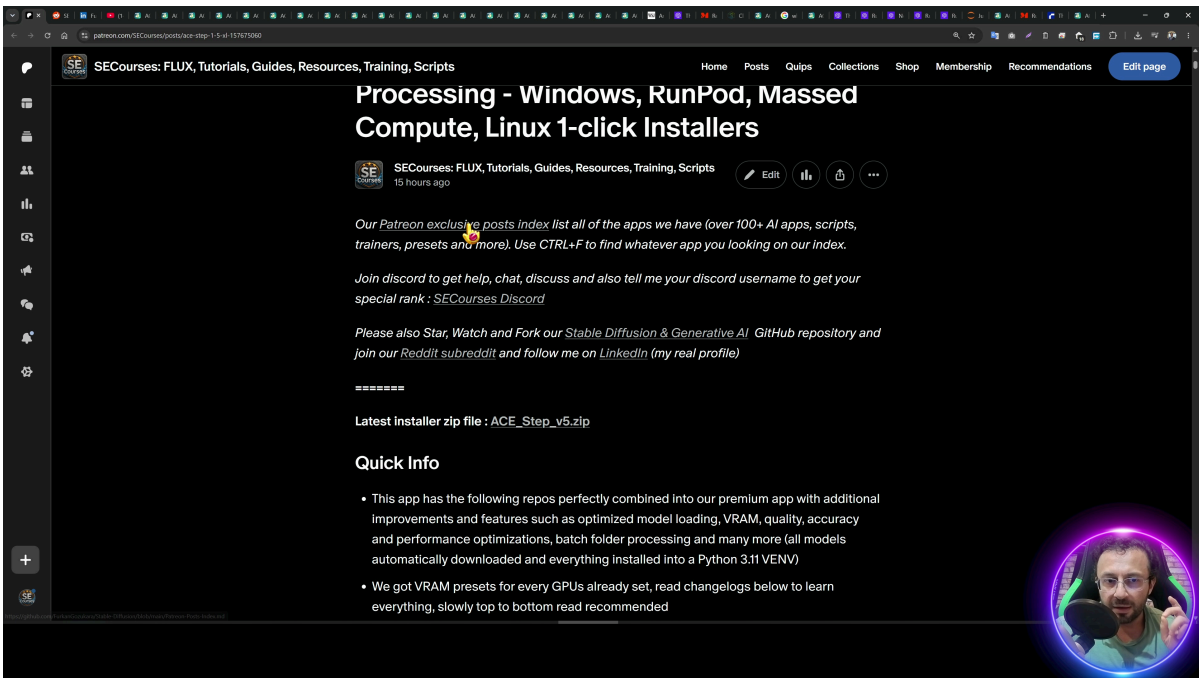
The SimplePod chapter uses the RunPod/SimplePod instruction file and shows a persistent-storage flow that resembles RunPod. The tutorial demonstrates starting, generating, monitoring, stopping, and resuming from the same storage volume.



SimplePod setup



SimplePod generation



SimplePod resume

1. Register, add credits, and create/use persistent storage as shown in the instruction file.
2. Open the template link, attach the storage volume, choose a GPU, and run the machine.
3. Use the JupyterLab or console link to run the installer/start commands from the workspace.
4. If the Gradio live page throws a first-click error, refresh or click again after the page is fully loaded.
5. Install `nvitop` when you want GPU/VRAM visibility: `pip install nvitop`, then run `nvitop`.
6. To resume, reuse the template link, attach the same volume, select a GPU, start the machine, and run the app start command again.
7. Stop or terminate compute and remove storage when finished to avoid unwanted billing.

13. Troubleshooting And Best Practices

- If Gradio errors after opening, refresh the browser or click again after the page finishes loading.
- If RunPod installer errors, rerun the installer. If the virtual environment is corrupt, delete only the virtual environment and run the installer again.
- If first generation is slow, wait for model load and monitor the terminal or `nvitop`.
- For Remix quality, use SFT when available, keep lyrics aligned with the source, generate until you get a good base, then lock the seed.
- For very large videos, use Disable upload preview in Audio Processing or supply a local path when available.
- Save presets and keep `generation_manifest.json` with outputs.
- For VRAM pressure, use GPU Optimization Preset, quantization, offload controls, smaller duration/count, lower SAM candidates, or a larger GPU.
- Stop/terminate cloud compute and delete storage volumes when the tutorial work is finished.